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(54) MULTI-CRITERIA BLOCKCHAIN PROTOCOL

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CPC ***G06Q 20/401*** (2013.01); ***G06Q 20/3825*** (2013.01); ***G06Q 20/407*** (2013.01); ***H04L 9/50*** (2022.05); *G06Q 2220/00* (2013.01)

(57) **ABSTRACT**

A computer-implemented method of generating a transaction for a blockchain, the transaction being for transferring an amount of a digital asset from a first party to a second party. The method comprises generating a first transaction comprising an output locking the amount of the digital asset, the output comprising an output script comprising a plurality of criterion components each requiring a respective input data item, and a plurality of counter script components. Each criterion component is associated with one of the counter script components. The output script is configured so as to, when executed alongside an input script of a second transaction, i) increment a counter each time a respective criterion component is satisfied by a respective input data item of the input script, and ii) to require the counter to increment to at least a predetermined number in order to be unlocked by the input script.

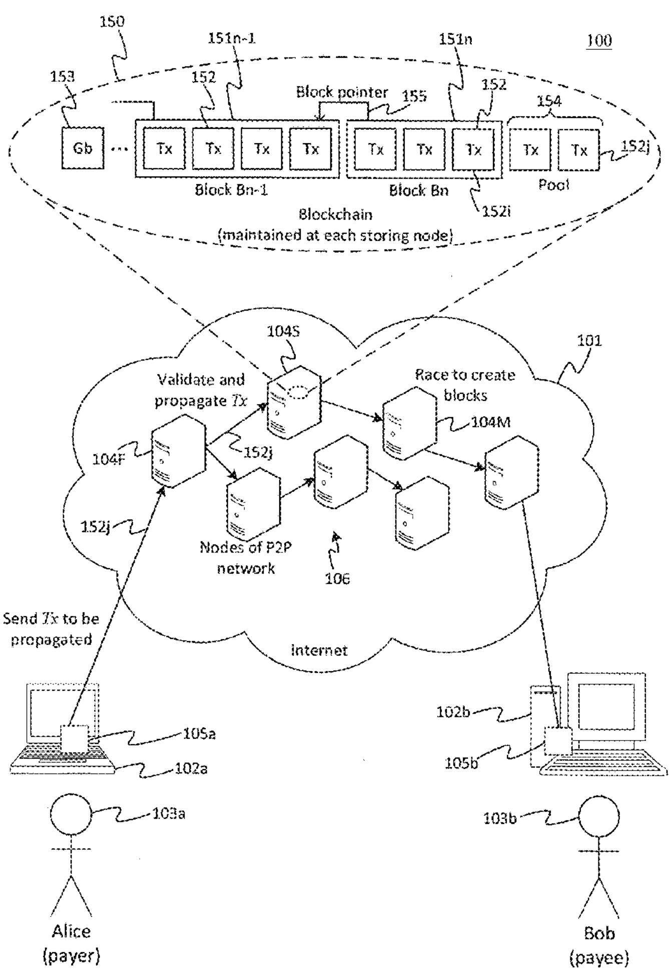


Figure 1

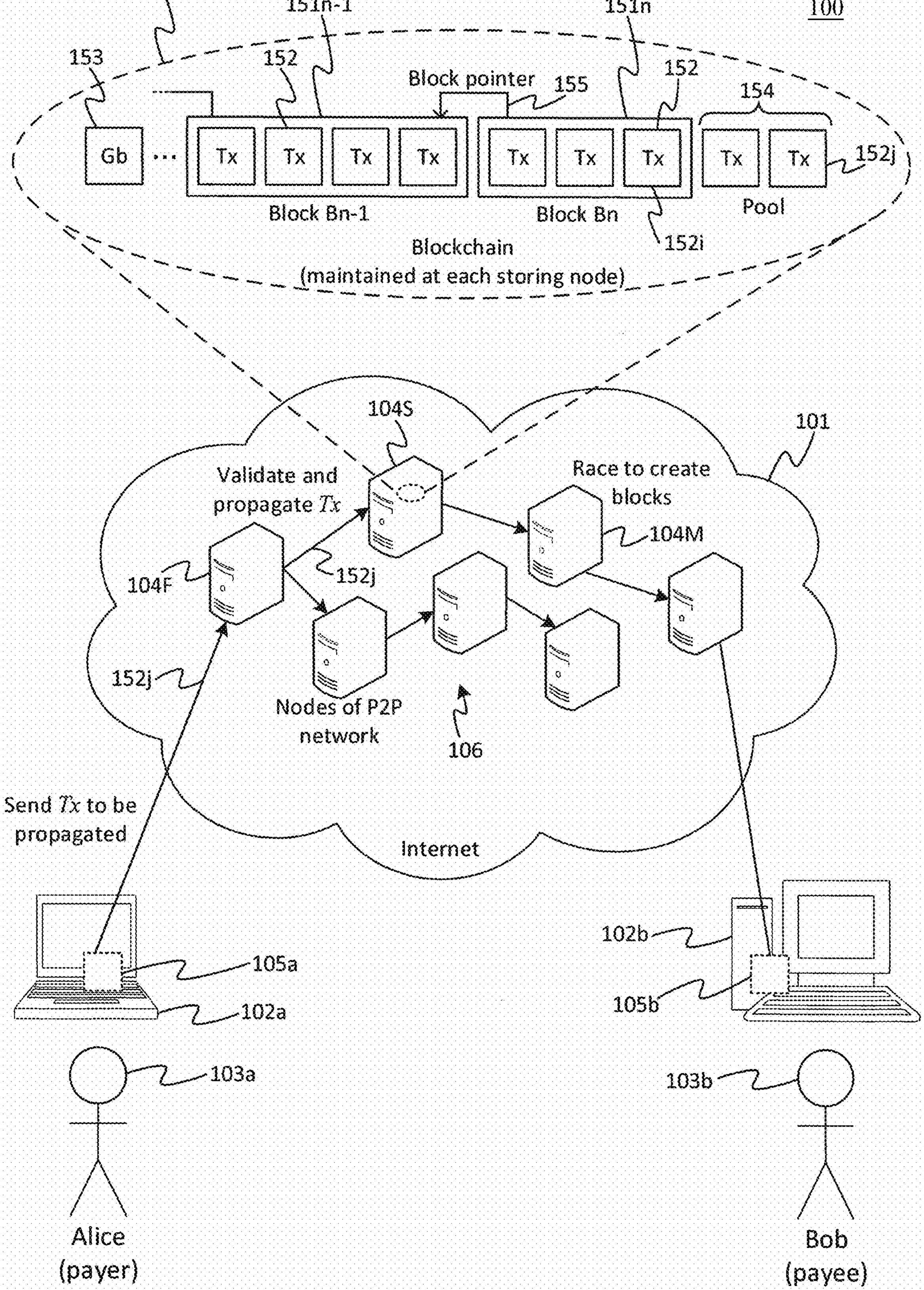

100
150
151n-1
151n
153
152
Block pointer
155
152
154
Gb
Tx
Tx
Tx
Tx
Tx
Tx
Tx
Tx
Tx
152j
Block Bn-1
Block Bn
Pool
152i
Blockchain
(maintained at each storing node)
104S
101
Validate and propagate Tx
Race to create blocks
104M
104F
152j
152j
Nodes of P2P network
106
Send Tx to be propagated
Internet
102b
105a
105b
102a
103a
103b
Alice
(payer)
Bob
(payee)

Figure 2

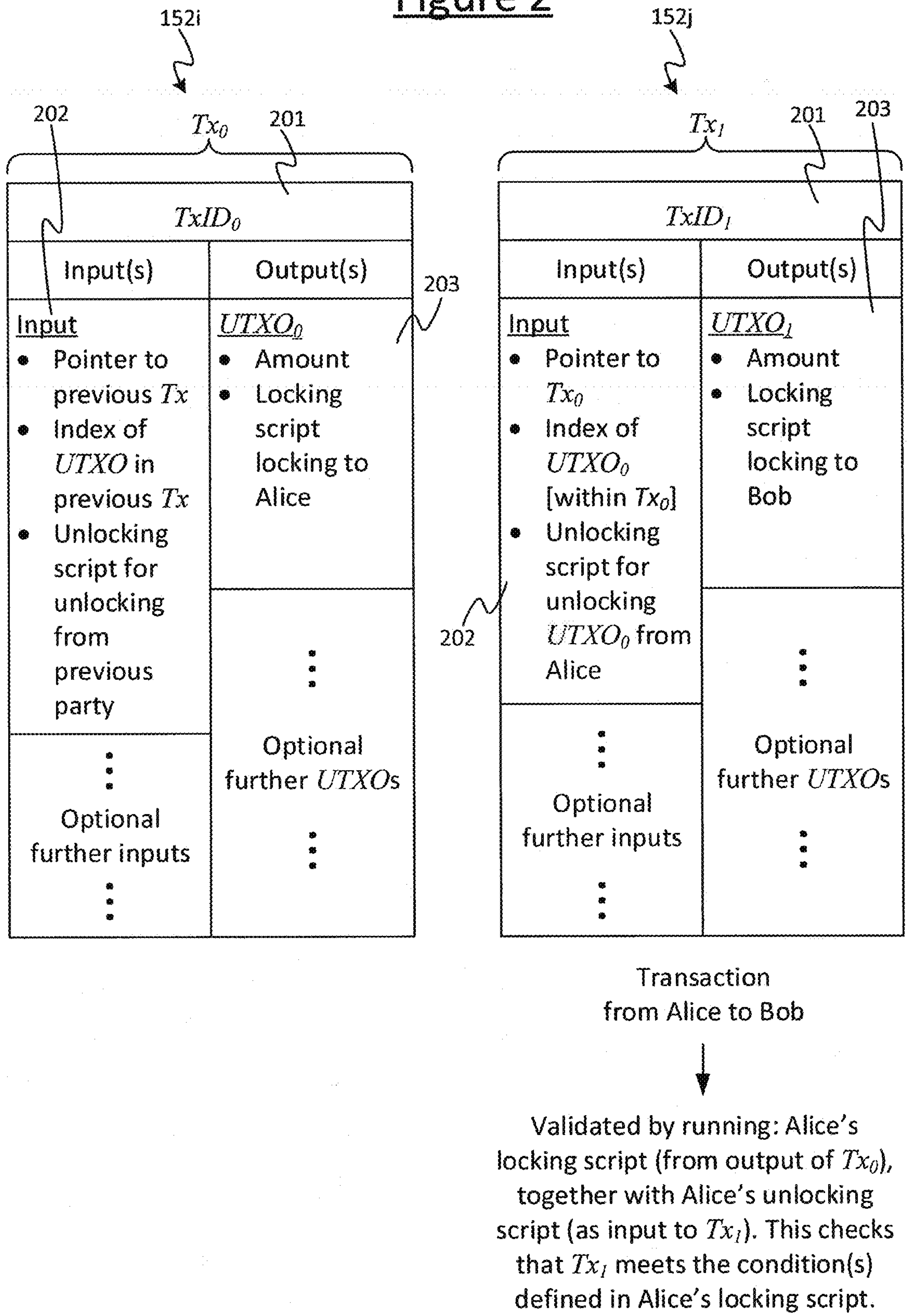

152i
152j
202
201
203
Tx0
TxID0
Input(s)
Output(s)
Input
Pointer to previous Tx
Index of UTXO in previous Tx
Unlocking script for unlocking from previous party
Optional further inputs
UTXO0
Amount
Locking script locking to Alice
Optional further UTXOs
Tx1
TxID1
Input
Pointer to Tx0
Index of UTXO0 [within Tx0]
Unlocking script for unlocking UTXO0 from Alice
Optional further inputs
UTXO1
Amount
Locking script locking to Bob
Optional further UTXOs
Transaction from Alice to Bob
Validated by running: Alice's locking script (from output of Tx0), together with Alice's unlocking script (as input to Tx1). This checks that Tx1 meets the condition(s) defined in Alice's locking script.

Figure 3

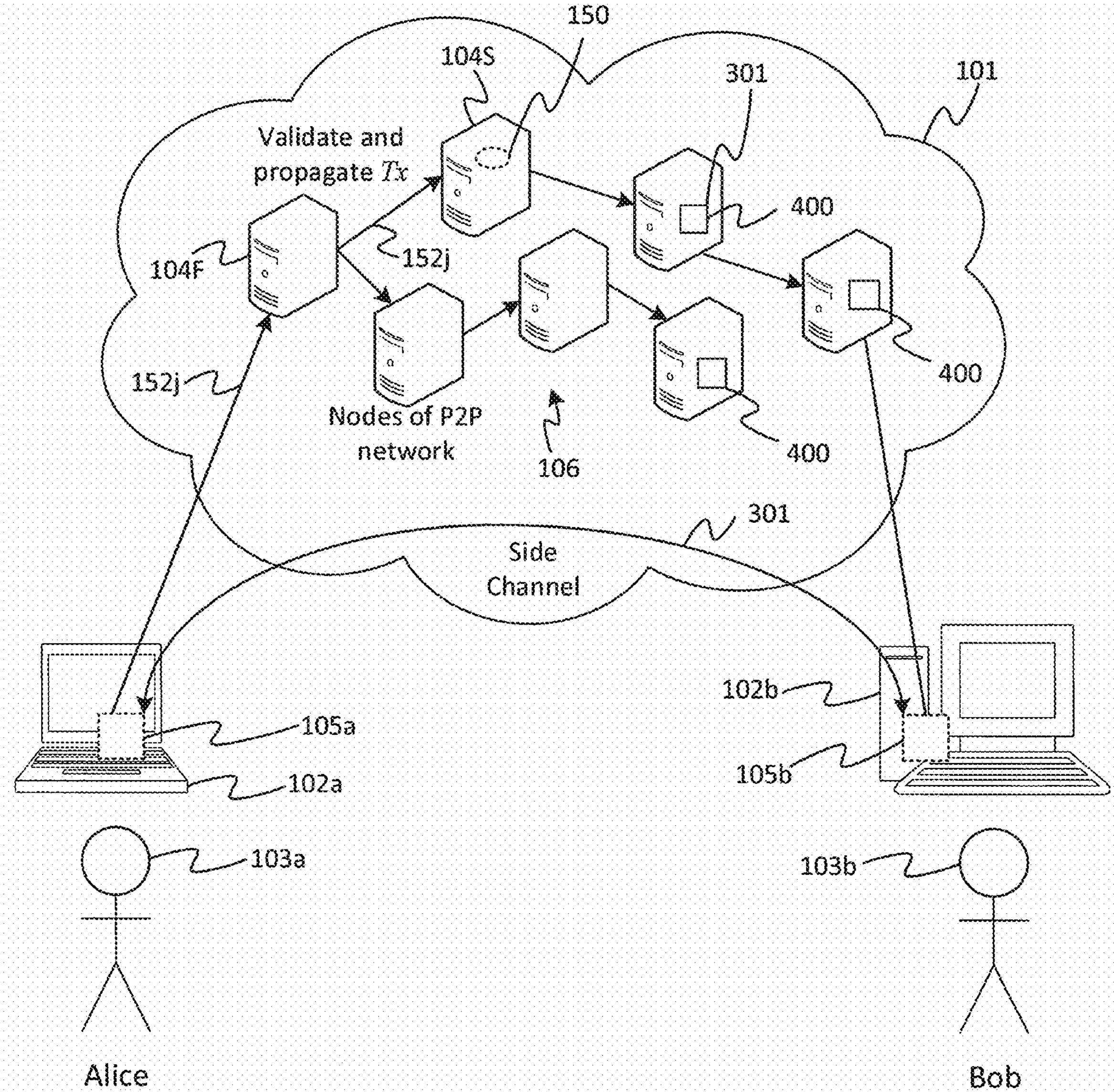

150
104S
301
101
Validate and propagate Tx
400
104F
152j
152j
400
Nodes of P2P network
106
400
301
Side Channel
102b
105a
105b
102a
103a
103b
Alice
Bob

Figure 4

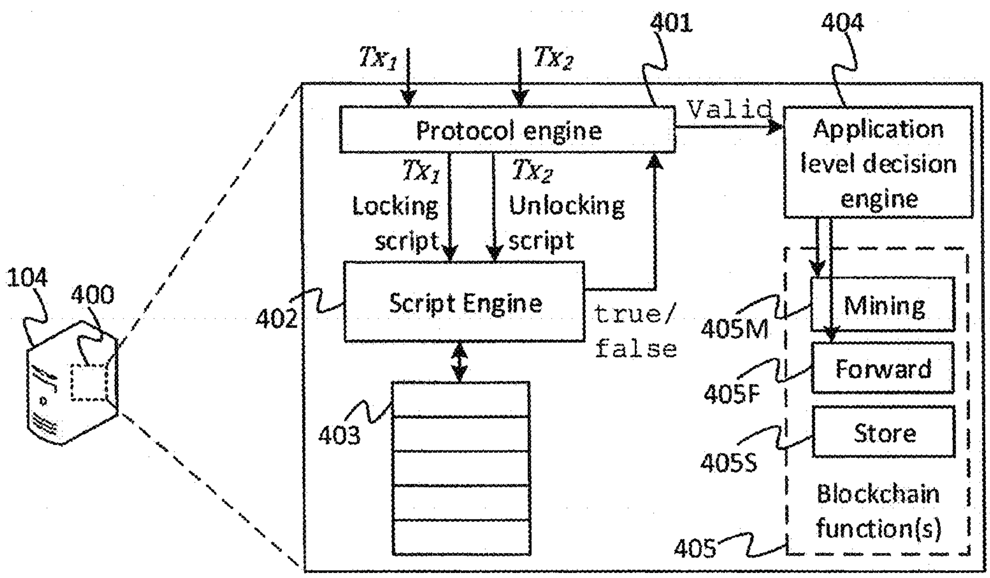


Figure 5

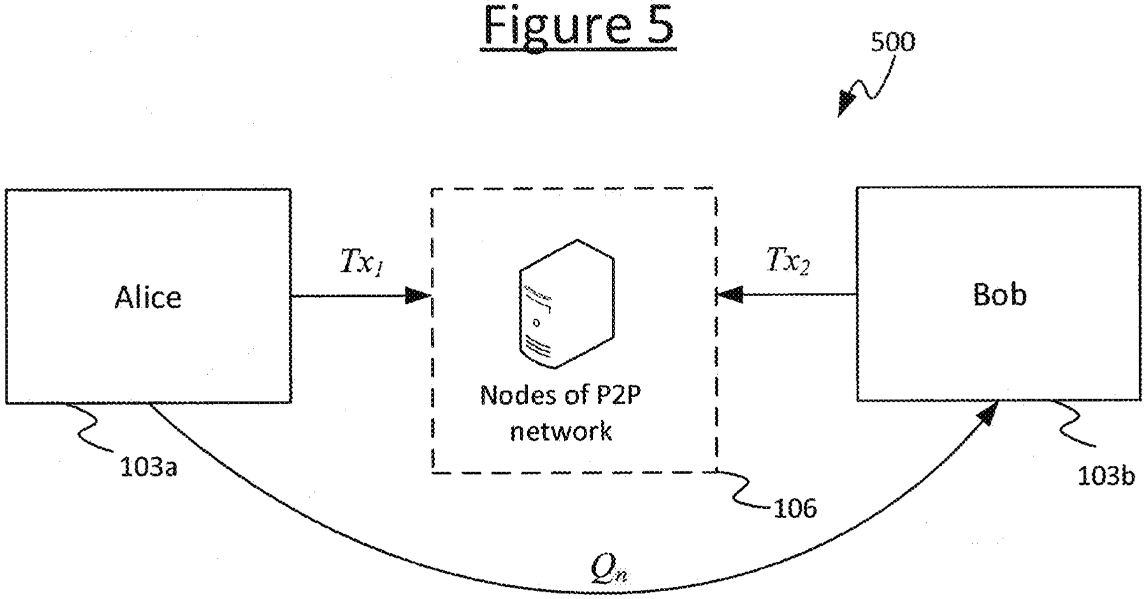


Figure 6

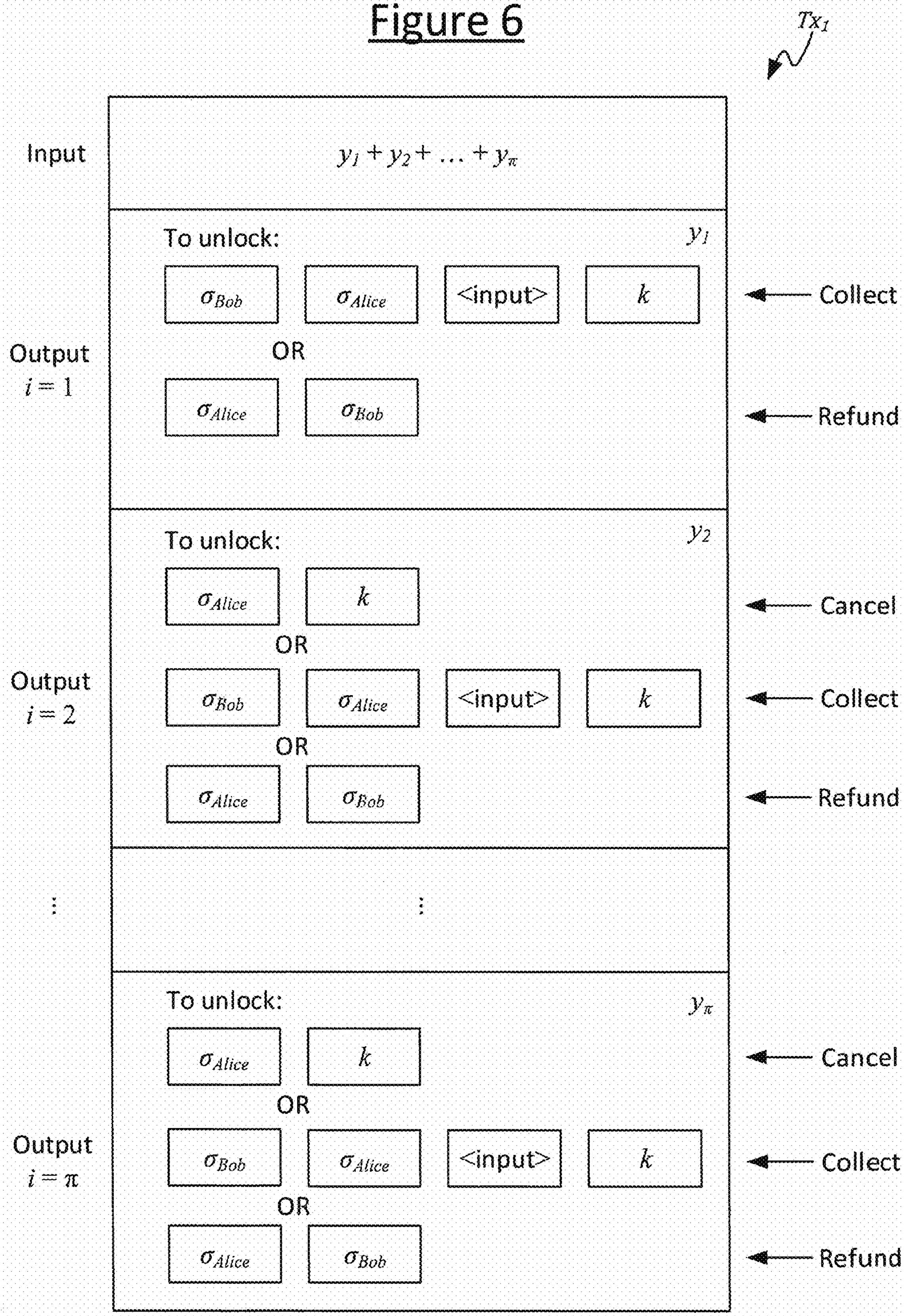

Tx1
Input
y1 + y2 + … + yπ
Output
i = 1
To unlock:
y1
σBob
σAlice
<input>
k
Collect
OR
σAlice
σBob
Refund
Output
i = 2
To unlock:
y2
σAlice
k
Cancel
OR
σBob
σAlice
<input>
k
Collect
OR
σAlice
σBob
Refund
Output
i = π
To unlock:
yπ
σAlice
k
Cancel
OR
σBob
σAlice
<input>
k
Collect
OR
σAlice
σBob
Refund

Figure 7

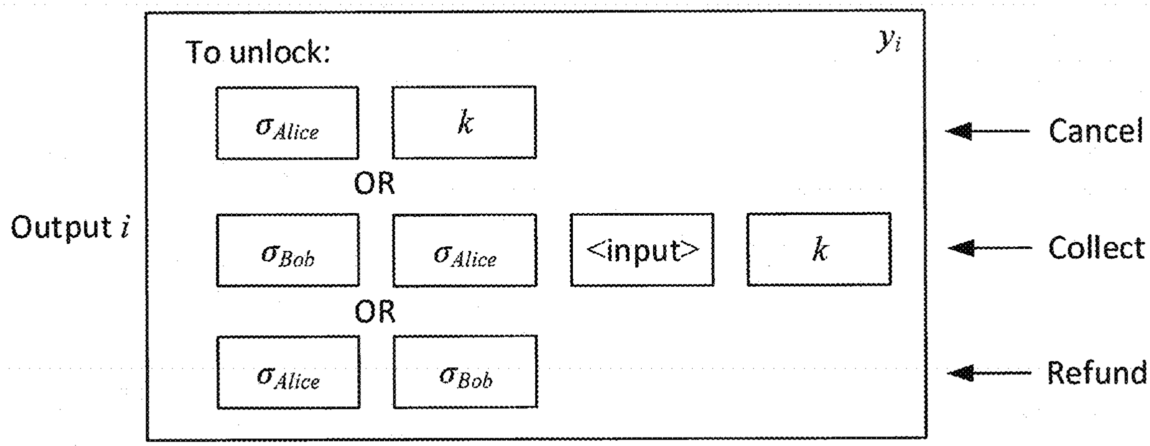


Figure 8

800

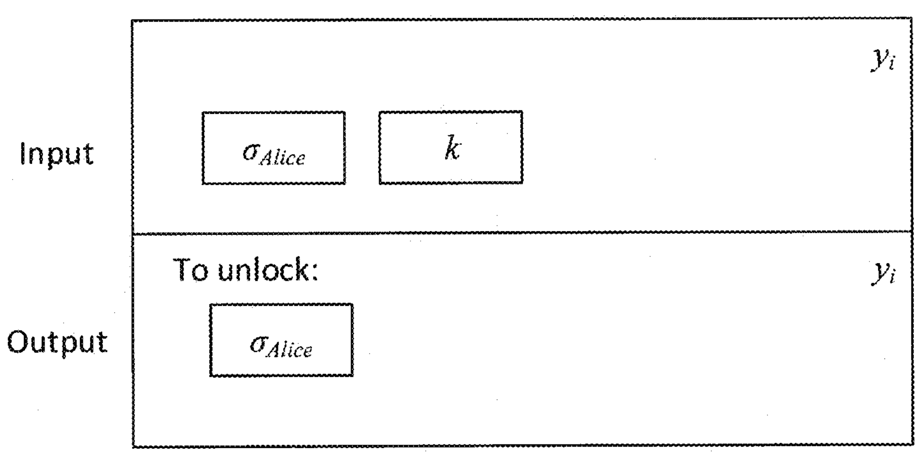


Figure 9

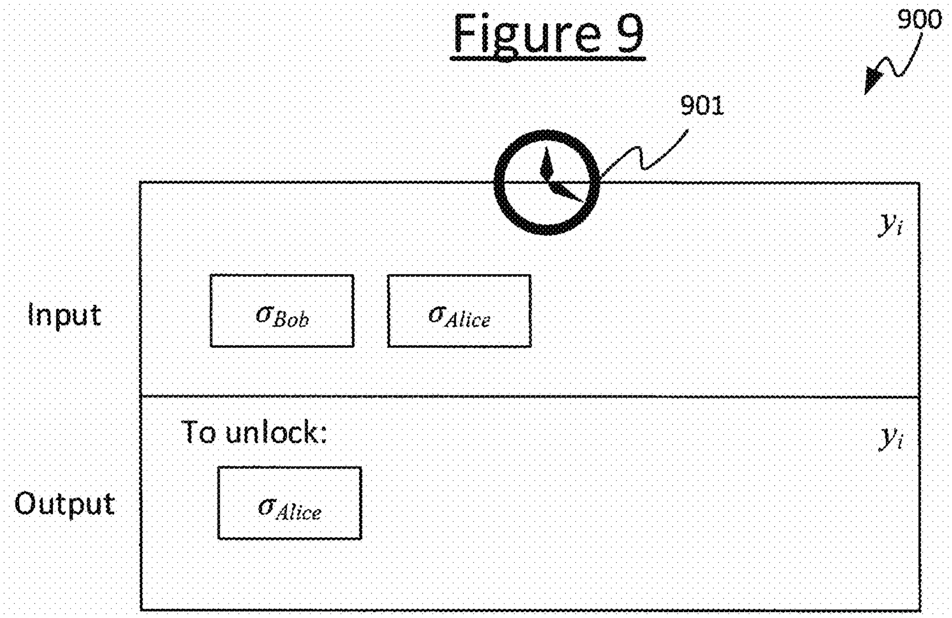


Figure 10

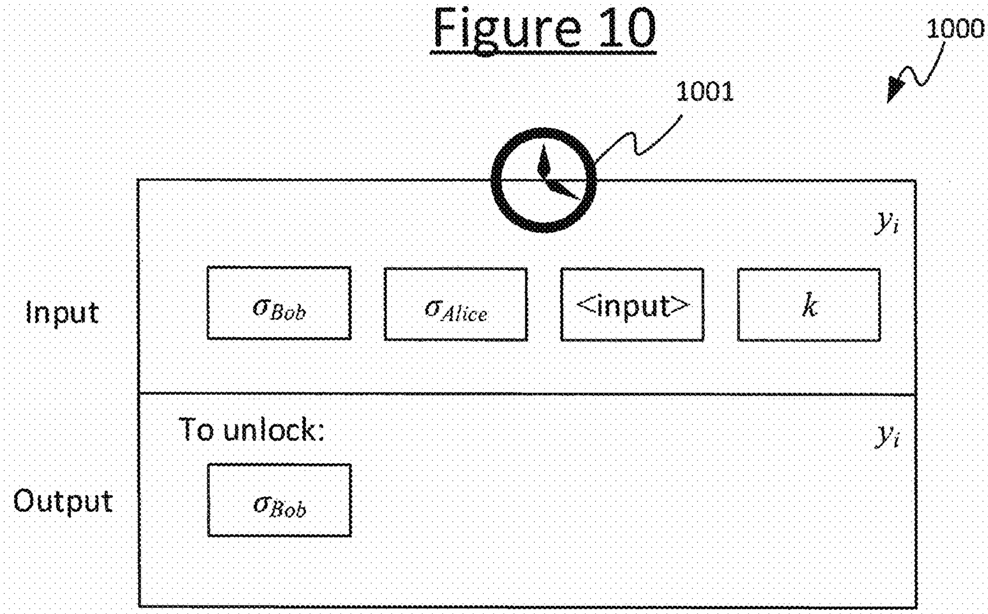


Figure 11

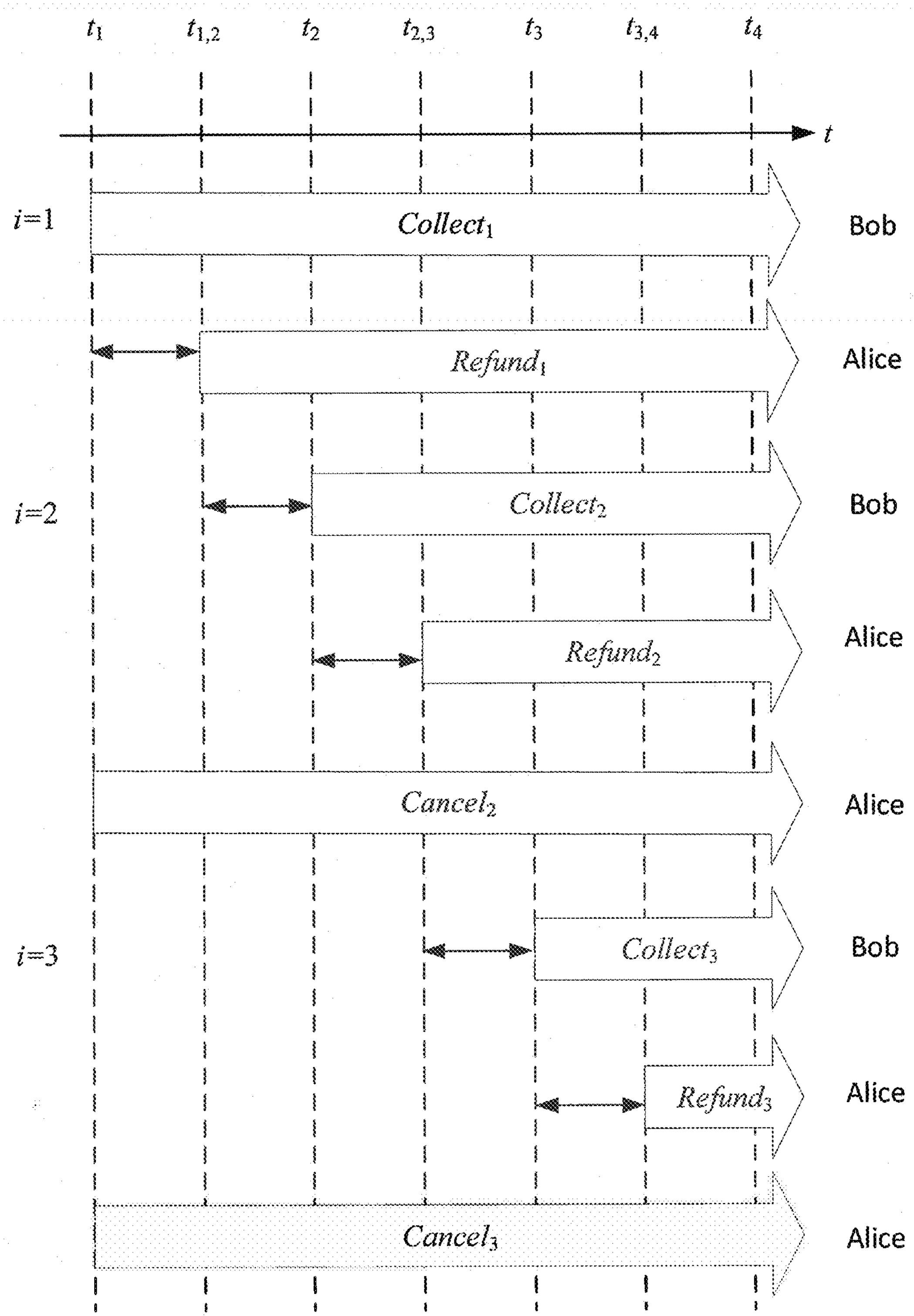
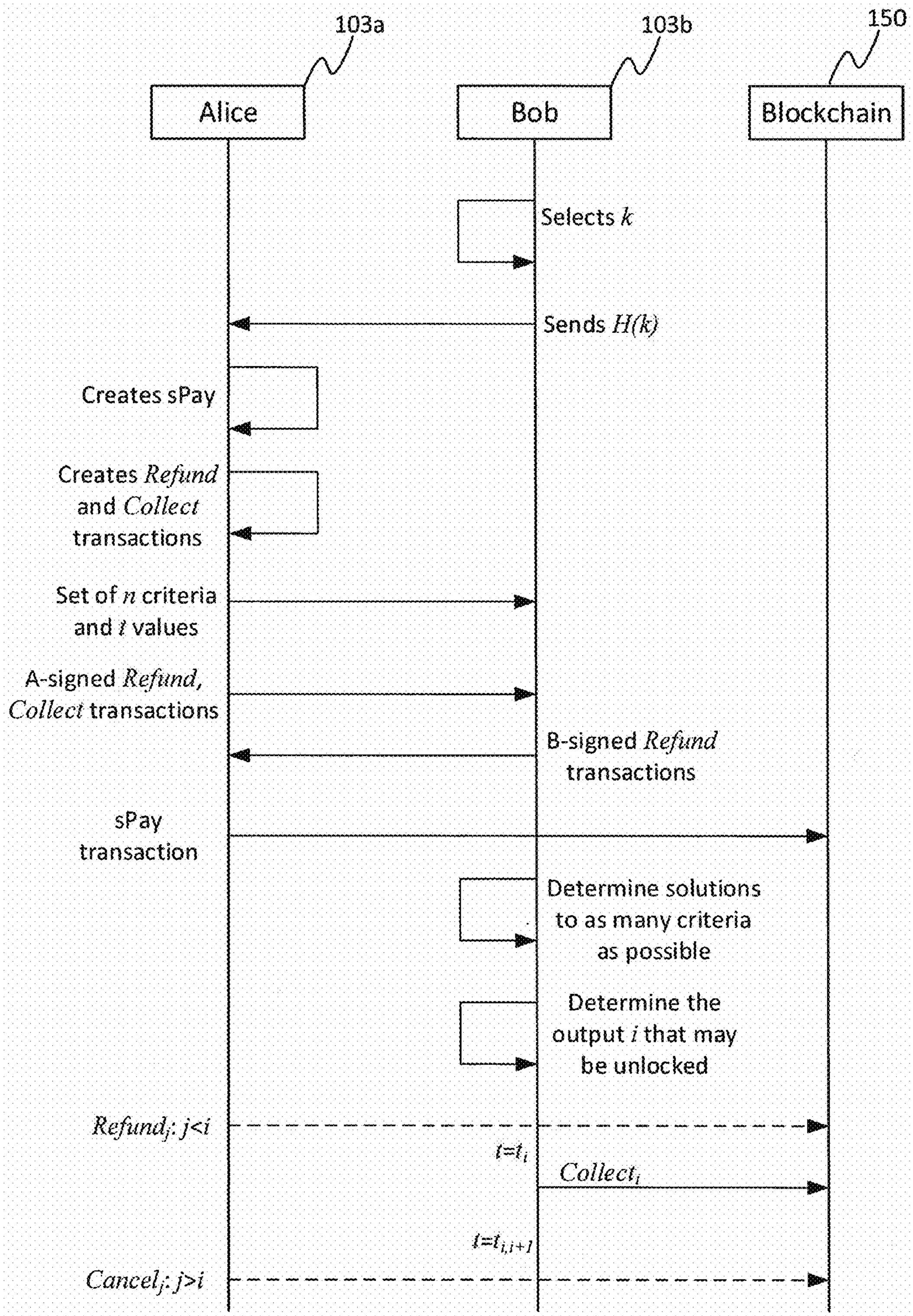

t_1
$t_{1,2}$
t_2
$t_{2,3}$
t_3
$t_{3,4}$
t_4
t
i=1
Collect$_1$
Bob
Refund$_1$
Alice
i=2
Collect$_2$
Bob
Refund$_2$
Alice
Cancel$_2$
Alice
i=3
Collect$_3$
Bob
Refund$_3$
Alice
Cancel$_3$
Alice

Figure 12



103a
103b
150
Alice
Bob
Blockchain
Selects k
Sends $H(k)$
Creates sPay
Creates *Refund* and *Collect* transactions
Set of n criteria and t values
A-signed *Refund*, *Collect* transactions
B-signed *Refund* transactions
sPay transaction
Determine solutions to as many criteria as possible
Determine the output i that may be unlocked
$Refund_j$: $j<i$
$t=t_i$
$Collect_i$
$t=t_{i,i+1}$
$Cancel_j$: $j>i$

Figure 13

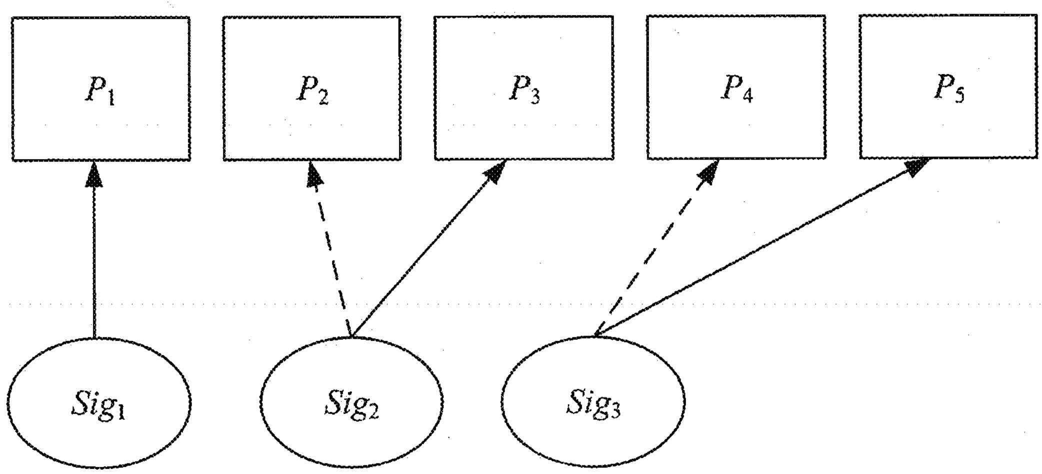
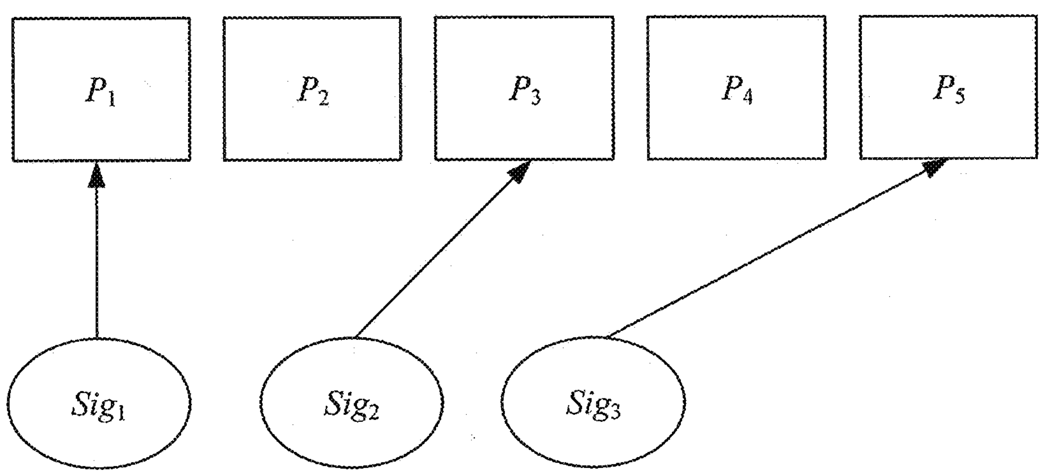


Figure 14



MULTI-CRITERIA BLOCKCHAIN PROTOCOL

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 17/760,728, filed on Mar. 15, 2022, which is the U.S. National Stage of International Application No. PCT/IB2020/057764 filed on Aug. 18, 2020, which claims the benefit of United Kingdom Patent Application No. 1913385.9, filed on Sep. 17, 2019, the contents of which are all incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to methods for and transactions for implementing a blockchain protocol which requires m-of-n criteria to be satisfied in order to unlock a transaction.

BACKGROUND

[0003] A blockchain refers to a form of distributed data structure, wherein a duplicate copy of the blockchain is maintained at each of a plurality of nodes in a peer-to-peer (P2P) network. The blockchain comprises a chain of blocks of data, wherein each block comprises one or more transactions. Each transaction may point back to a preceding transaction in a sequence which may span one or more blocks. Transactions can be submitted to the network to be included in new blocks by a process known as "mining", which involves each of a plurality of mining nodes competing to perform "proof-of-work", i.e. solving a cryptographic puzzle based on a pool of the pending transactions waiting to be included in blocks.

[0004] Conventionally the transactions in the blockchain are used to convey a digital asset, i.e. data acting as a store of value. However, a blockchain can also be exploited in order to layer additional functionality on top of the blockchain. For instance, blockchain protocols may allow for storage of additional user data in an output of a transaction. Modern blockchains are increasing the maximum data capacity that can be stored within a single transaction, enabling more complex data to be incorporated. For instance, this may be used to store an electronic document in the blockchain, or even audio or video data.

[0005] Each node in the network can have any one, two or all of three roles: forwarding, mining and storage. Forwarding nodes propagate transactions throughout the nodes of the network. Mining nodes perform the mining of transactions into blocks. Storage nodes each store their own copy of the mined blocks of the blockchain. In order to have a transaction recorded in the blockchain, a party sends the transaction to one of the nodes of the network to be propagated. Mining nodes which receive the transaction may race to mine the transaction into a new block. Each node is configured to respect the same node protocol, which will include one or more conditions for a transaction to be valid. Invalid transactions will not be propagated nor mined into blocks. Assuming the transaction is validated and thereby accepted onto the blockchain, the additional user data will thus remain stored at each of the nodes in the P2P network as an immutable public record.

[0006] Some blockchain protocols use a scripting language that gives parties the ability to create advanced criteria under which a potential recipient may spend the output of a transaction. Some scripting languages include a function (sometimes referred to as an opcode) which checks whether a sufficient number of valid digital signatures are present in the input of a transaction. If the required m-of-n signatures are available, then the output can be spent.

SUMMARY

[0007] While useful, m-of-n signature checks are limited to being a check only on whether the threshold of digital signatures requested has been met. However, there may be scenarios where thresholds for other criteria that are not necessarily signatures may be required. As an example, a potential recipient may be asked to produce 'at least a specified number of hash pre-images' in order to spend the output of a transaction.

[0008] According to one aspect disclosed herein, there is provided a computer-implemented method of generating a transaction for a blockchain, the transaction being for transferring an amount of a digital asset from a first party to a second party; the method being performed by the first party and comprising: generating a first transaction comprising an output locking the amount of the digital asset, the output comprising an output script comprising a plurality of criterion components each requiring a respective input data item, and a plurality of counter script components, each criterion component being associated with one of the counter script components, and wherein the output script is configured so as to, when executed alongside an input script of a second transaction, i) increment a counter each time a respective criterion component is satisfied by a respective input data item of the input script, and ii) to require the counter to increment to at least a predetermined number in order to be unlocked by the input script.

[0009] Thus a binary threshold protocol (BTP) is provided where a threshold restriction can be applied to any type of criteria. The output script condition defines multiple criterions which can be satisfied with input data items contained in a transaction trying to spend the digital asset. Each time a criterion is satisfied, a counter is incremented (e.g. by one). If the counter has reached the predetermined number (the binary threshold) by the time all of the input data items have been used up, the first script condition will be unlocked. If the input script satisfies any other conditions of the protocol (e.g. the transaction is a valid transaction), the amount of the digital asset will be transferred to the second party). The protocol is binary in the sense that, if the threshold is reached the digital asset can be transferred (e.g. spent), whereas if the threshold is not reached the digital asset cannot be transferred.

[0010] While the BTP provides advantages, it may be desirable in some cases that the amount of the digital asset transferred to the second party is based on the number of criteria which are satisfied (e.g. more of the digital asset is transferred if more criteria are satisfied). For example, it may be desirable to transfer a first amount if an intended recipient can provide 1-of-n pre-images, a larger amount if 2-of-n pre-images are provided, and an even larger amount of 3-of-n pre-images are provided.

[0011] According to another aspect disclosed herein, there is provided a computer-implemented method of generating a transaction for a blockchain, the transaction being for transferring an amount of a digital asset from a first party to a second party, the first and second parties each being associated with the blockchain; the method being performed by the first party and comprising: generating a first transaction comprising a plurality of outputs, each output locking a respective amount of the digital asset and comprising a respective first script condition, wherein each first script condition comprises a plurality of criterion components each requiring a respective input data item, and a plurality of counter components, each criterion component being associated with one of the counter components, and wherein each first output script condition is configured so as to, when executed alongside an input script of a second transaction, i) increment a counter each time a respective criterion component is satisfied by a respective input data item of the input script, and ii) require the counter to increment to at least a respective predetermined number in order to be unlocked by that input script, and wherein one, some or all of the first script conditions require the counter to increment to a different predetermined number.

[0012] Like the BTP, in this method a spectrum threshold protocol (STP) is provided where a threshold restriction can be applied to any type of criteria. The first transaction has multiple outputs, each of which would transfer an amount of digital asset if unlocked. For example, each output may transfer a different amount of digital asset, with a first output transferring a first amount, a second output transferring a different (e.g. larger) amount, and so on. Each output has a first output script condition having multiple criteria which can be satisfied with input data items contained in a transaction trying to spend the digital asset. Each time a criterion is satisfied, a counter is incremented (e.g. by one). Some of the first output script conditions require the counter to reach a different predetermined number in order to be unlocked. That is, at least one output script will be unlocked with a lower number of satisfied criteria compared to another output script. This creates a spectrum threshold. For example, satisfying a smaller number of criteria may unlock an output which locks a smaller amount of digital asset whilst satisfying a larger number of criteria may unlock an output which locks a larger amount of digital asset.

[0013] Each individual output of a first transaction following the STP is similar to an output of a first transaction following the BTP protocol, with the predetermined number (threshold) required by the STP outputs varying amongst some or all of the outputs. For each threshold protocol, a set of transactions are employed to enable a recipient (second party) to access the digital asset based on the number of criteria that the recipient satisfies. The STP is applicable for cases where the amount of digital asset (reward) is proportional to the number of criteria the recipient is able to satisfy, and also for cases where the reward is not proportional to the number of satisfied criteria. For instance, there may be a cap on the reward where above a certain threshold, satisfying more criteria would not increase the obtainable reward.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] To assist understanding of embodiments of the present disclosure and to show how such embodiments may be put into effect, reference is made, by way of example only, to the accompanying drawings in which:

[0015] FIG. **1** is a schematic block diagram of a system for implementing a blockchain;

[0016] FIG. **2** schematically illustrates some examples of transactions which may be recorded in a blockchain;

[0017] FIG. **3** is a schematic block diagram of another system for implementing a blockchain;

[0018] FIG. **4** is a schematic block diagram of a piece of node software for processing transactions in accordance with a node protocol of an output-based model;

[0019] FIG. **5** is a simplified version of FIG. **1** and schematically represents a system for implementing generic criteria threshold blockchain transactions;

[0020] FIG. **6** is a schematic representation of a Spectrum Pay transaction;

[0021] FIG. **7** is a schematic representation of an output of a Spectrum Pay transaction

[0022] FIG. **8** is a schematic representation of a Cancel transaction;

[0023] FIG. **9** is a schematic representation of a Refund transaction;

[0024] FIG. **10** is a schematic representation of a Collect transaction;

[0025] FIG. **11** schematically illustrates an example sequence of time-staggered transactions;

[0026] FIG. **12** also schematically illustrates a timeline showing an example sequence of steps taken by different parties to implement a generic criteria threshold blockchain protocol;

[0027] FIG. **13** is a schematic representation of an implementation for checking m-of-n signatures using OP_CHECKMULTISIG blockchain protocol; and

[0028] FIG. **14** is a schematic representation of an implementation for checking m-of-n signatures using a generic criteria threshold blockchain protocol.

DETAILED DESCRIPTION OF EMBODIMENTS

Example System Overview

[0029] FIG. **1** shows an example system **100** for implementing a blockchain **150** generally. The system **100** comprises a packet-switched network **101**, typically a wide-area internetwork such as the Internet. The packet-switched network **101** comprises a plurality of nodes **104** arranged to form a peer-to-peer (P2P) overlay network **106** within the packet-switched network **101**. Each node **104** comprises computer equipment of a peers, with different ones of the nodes **104** belonging to different peers. Each node **104** comprises processing apparatus comprising one or more processors, e.g. one or more central processing units (CPUs), accelerator processors, application specific processors and/or field programmable gate arrays (FPGAs). Each node also comprises memory, i.e. computer-readable storage in the form of a non-transitory computer-readable medium or media. The memory may comprise one or more memory units employing one or more memory media, e.g. a magnetic medium such as a hard disk; an electronic medium such as a solid-state drive (SSD), flash memory or EEPROM; and/or an optical medium such as an optical disk drive.

[0030] The blockchain **150** comprises a chain of blocks of data **151**, wherein a respective copy of the blockchain **150** is maintained at each of a plurality of nodes in the P2P network **160**. Each block **151** in the chain comprises one or more transactions **152**, wherein a transaction in this context refers to a kind of data structure. The nature of the data structure will depend on the type of transaction protocol used as part of a transaction model or scheme. A given blockchain will typically use one particular transaction protocol throughout. In one common type of transaction protocol, the data structure of each transaction **152** comprises at least one input and at least one output. Each output specifies an amount representing a quantity of a digital asset belonging to a user **103** to whom the output is cryptographically locked (requiring a signature of that user in order to be unlocked and thereby redeemed or spent). Each input points back to the output of a preceding transaction **152**, thereby linking the transactions.

[0031] At least some of the nodes **104** take on the role of forwarding nodes **104**F which forward and thereby propagate transactions **152**. At least some of the nodes **104** take on the role of miners **104**M which mine blocks **151**. At least some of the nodes **104** take on the role of storage nodes **104**S (sometimes also called "full-copy" nodes), each of which stores a respective copy of the same blockchain **150** in their respective memory. Each miner node **104**M also maintains a pool **154** of transactions **152** waiting to be mined into blocks **151**. A given node **104** may be a forwarding node **104**, miner **104**M, storage node **104**S or any combination of two or all of these.

[0032] In a given present transaction **152***j*, the (or each) input comprises a pointer referencing the output of a preceding transaction **152***i* in the sequence of transactions, specifying that this output is to be redeemed or "spent" in the present transaction **152***j*. In general, the preceding transaction could be any transaction in the pool **154** or any block **151**. The preceding transaction **152***i* need not necessarily exist at the time the present transaction **152***j* is created or even sent to the network **106**, though the preceding transaction **152***i* will need to exist and be validated in order for the present transaction to be valid. Hence "preceding" herein refers to a predecessor in a logical sequence linked by pointers, not necessarily the time of creation or sending in a temporal sequence, and hence it does not necessarily exclude that the transactions **152***i*, **152***j* be created or sent out-of-order (see discussion below on orphan transactions). The preceding transaction **152***i* could equally be called the antecedent or predecessor transaction.

[0033] The input of the present transaction **152***j* also comprises the signature of the user **103***a* to whom the output of the preceding transaction **152***i* is locked. In turn, the output of the present transaction **152***j* can be cryptographically locked to a new user **103***b*. The present transaction **152***j* can thus transfer the amount defined in the input of the preceding transaction **152***i* to the new user **103***b* as defined in the output of the present transaction **152***j*. In some cases, a transaction **152** may have multiple outputs to split the input amount between multiple users (one of whom could be the original user **103***a* in order to give change). In some cases, a transaction can also have multiple inputs to gather together the amounts from multiple outputs of one or more preceding transactions, and redistribute to one or more outputs of the current transaction.

[0034] The above may be referred to as an "output-based" transaction protocol, sometimes also referred to as an unspent transaction output (UTXO) type protocol (where the outputs are referred to as UTXOs). A user's total balance is not defined in any one number stored in the blockchain, and instead the user needs a special "wallet" application **105** to collate the values of all the UTXOs of that user which are scattered throughout many different transactions **152** in the blockchain **151**.

[0035] An alternative type of transaction protocol may be referred to as an "account-based" protocol, as part of an account-based transaction model. In the account-based case, each transaction does not define the amount to be transferred by referring back to the UTXO of a preceding transaction in a sequence of past transactions, but rather by reference to an absolute account balance. The current state of all accounts is stored by the miners separate to the blockchain and is updated constantly. In such a system, transactions are ordered using a running transaction tally of the account (also called the "position"). This value is signed by the sender as part of their cryptographic signature and is hashed as part of the transaction reference calculation. In addition, an optional data field may also be signed the transaction. This data field may point back to a previous transaction, for example if the previous transaction ID is included in the data field.

[0036] With either type of transaction protocol, when a user **103** wishes to enact a new transaction **152***j*, then he/she sends the new transaction from his/her computer terminal **102** to one of the nodes **104** of the P2P network **106** (which nowadays are typically servers or data centres, but could in principle be other user terminals). This node **104** checks whether the transaction is valid according to a node protocol which is applied at each of the nodes **104**.

[0037] The details of the node protocol will correspond to the type of transaction protocol being used in the blockchain **150** in question, together forming the overall transaction model. The node protocol typically requires the node **104** to check that the cryptographic signature in the new transaction **152***j* matches the expected signature, which depends on the previous transaction **152***i* in an ordered sequence of transactions **152**. In an output-based case, this may comprise checking that the cryptographic signature of the user included in the input of the new transaction **152***j* matches a condition defined in the output of the preceding transaction **152***i* which the new transaction spends, wherein this condition typically comprises at least checking that the cryptographic signature in the input of the new transaction **152***j* unlocks the output of the previous transaction **152***i* to which the input of the new transaction points. In some transaction protocols the condition may be at least partially defined by a custom script included in the input and/or output. Alternatively, it could simply be a fixed by the node protocol alone, or it could be due to a combination of these. Either way, if the new transaction **152***j* is valid, the current node forwards it to one or more others of the nodes **104** in the P2P network **106**. At least some of these nodes **104** also act as forwarding nodes **104**F, applying the same test according to the same node protocol, and so forward the new transaction **152***j* on to one or more further nodes **104**, and so forth. In this way the new transaction is propagated throughout the network of nodes **104**.

[0038] In an output-based model, the definition of whether a given output (e.g. UTXO) is spent is whether it has yet been validly redeemed by the input of another, onward transaction **152***j* according to the node protocol. Another condition for a transaction to be valid is that the output of the preceding transition **152***i* which it attempts to spend or redeem has not already been spent/redeemed by another valid transaction. Again, if not valid, the transaction **152***j* will not be propagated or recorded in the blockchain. This guards against double-spending whereby the spender tries to spend the output of the same transaction more than once. An account-based model on the other hand guards against double-spending by maintaining an account balance.

Because again there is a defined order of transactions, the account balance has a single defined state at any one time.

[0039] In addition to validation, at least some of the nodes **104**M also race to be the first to create blocks of transactions in a process known as mining, which is underpinned by "proof of work". At a mining node **104**M, new transactions are added to a pool of valid transactions that have not yet appeared in a block. The miners then race to assemble a new valid block **151** of transactions **152** from the pool of transactions **154** by attempting to solve a cryptographic puzzle. Typically, this comprises searching for a "nonce" value such that when the nonce is concatenated with the pool of transactions **154** and hashed, then the output of the hash meets a predetermined condition. E.g. the predetermined condition may be that the output of the hash has a certain predefined number of leading zeros. A property of a hash function is that it has an unpredictable output with respect to its input. Therefore, this search can only be performed by brute force, thus consuming a substantive amount of processing resource at each node **104**M that is trying to solve the puzzle.

[0040] The first miner node **104**M to solve the puzzle announces this to the network **106**, providing the solution as proof which can then be easily checked by the other nodes **104** in the network (once given the solution to a hash it is straightforward to check that it causes the output of the hash to meet the condition). The pool of transactions **154** for which the winner solved the puzzle then becomes recorded as a new block **151** in the blockchain **150** by at least some of the nodes **104** acting as storage nodes **104**S, based on having checked the winner's announced solution at each such node. A block pointer **155** is also assigned to the new block **151***n* pointing back to the previously created block **151***n*-**1** in the chain. The proof-of-work helps reduce the risk of double spending since it takes a large amount of effort to create a new block **151**, and as any block containing a double spend is likely to be rejected by other nodes **104**, mining nodes **104**M are incentivised not to allow double spends to be included in their blocks. Once created, the block **151** cannot be modified since it is recognized and maintained at each of the storing nodes **104**S in the P2P network **106** according to the same protocol. The block pointer **155** also imposes a sequential order to the blocks **151**. Since the transactions **152** are recorded in the ordered blocks at each storage node **104**S in a P2P network **106**, this therefore provides an immutable public ledger of the transactions.

[0041] Note that different miners **104**M racing to solve the puzzle at any given time may be doing so based on different snapshots of the unmined transaction pool **154** at any given time, depending on when they started searching for a solution. Whoever solves their respective puzzle first defines which transactions **152** are included in the next new block **151***n*, and the current pool **154** of unmined transactions is updated. The miners **104**M then continue to race to create a block from the newly defined outstanding pool **154**, and so forth. A protocol also exists for resolving any "fork" that may arise, which is where two miners **104**M solve their puzzle within a very short time of one another such that a conflicting view of the blockchain gets propagated. In short, whichever prong of the fork grows the longest becomes the definitive blockchain **150**.

[0042] In most blockchains the winning miner **104**M is automatically rewarded with a special kind of new transaction which creates a new quantity of the digital asset out of nowhere (as opposed to normal transactions which transfer an amount of the digital asset from one user to another). Hence the winning node is said to have "mined" a quantity of the digital asset. This special type of transaction is sometime referred to as a "generation" transaction. It automatically forms part of the new block **151***n*. This reward gives an incentive for the miners **104**M to participate in the proof-of-work race. Often a regular (non-generation) transaction **152** will also specify an additional transaction fee in one of its outputs, to further reward the winning miner **104**M that created the block **151***n* in which that transaction was included.

[0043] Due to the computational resource involved in mining, typically at least each of the miner nodes **104**M takes the form of a server comprising one or more physical server units, or even whole a data centre. Each forwarding node **104**M and/or storage node **104**S may also take the form of a server or data centre. However, in principle any given node **104** could take the form of a user terminal or a group of user terminals networked together.

[0044] The memory of each node **104** stores software configured to run on the processing apparatus of the node **104** in order to perform its respective role or roles and handle transactions **152** in accordance with the node protocol. It will be understood that any action attributed herein to a node **104** may be performed by the software run on the processing apparatus of the respective computer equipment. Also, the term "blockchain" as used herein is a generic term that refers to the kind of technology in general, and does not limit to any particular proprietary blockchain, protocol or service.

[0045] Also connected to the network **101** is the computer equipment **102** of each of a plurality of parties **103** in the role of consuming users. These act as payers and payees in transactions but do not necessarily participate in mining or propagating transactions on behalf of other parties. They do not necessarily run the mining protocol. Two parties **103** and their respective equipment **102** are shown for illustrative purposes: a first party **103***a* and his/her respective computer equipment **102***a*, and a second party **103***b* and his/her respective computer equipment **102***b*. It will be understood that many more such parties **103** and their respective computer equipment **102** may be present and participating in the system, but for convenience they are not illustrated. Each party **103** may be an individual or an organization. Purely by way of illustration the first party **103***a* is referred to herein as Alice and the second party **103***b* is referred to as Bob, but it will be appreciated that this is not limiting and any reference herein to Alice or Bob may be replaced with "first party" and "second party" respectively.

[0046] The computer equipment **102** of each party **103** comprises respective processing apparatus comprising one or more processors, e.g. one or more CPUs, GPUs, other accelerator processors, application specific processors, and/or FPGAs. The computer equipment **102** of each party **103** further comprises memory, i.e. computer-readable storage in the form of a non-transitory computer-readable medium or media. This memory may comprise one or more memory units employing one or more memory media, e.g. a magnetic medium such as hard disk; an electronic medium such as an SSD, flash memory or EEPROM; and/or an optical medium such as an optical disc drive. The memory on the computer equipment **102** of each party **103** stores software comprising a respective instance of at least one client application **105**

arranged to run on the processing apparatus. It will be understood that any action attributed herein to a given party **103** may be performed using the software run on the processing apparatus of the respective computer equipment **102**. The computer equipment **102** of each party **103** comprises at least one user terminal, e.g. a desktop or laptop computer, a tablet, a smartphone, or a wearable device such as a smartwatch. The computer equipment **102** of a given party **103** may also comprise one or more other networked resources, such as cloud computing resources accessed via the user terminal.

[0047] The client application or software **105** may be initially provided to the computer equipment **102** of any given party **103** on suitable computer-readable storage medium or media, e.g. downloaded from a server, or provided on a removable storage device such as a removable SSD, flash memory key, removable EEPROM, removable magnetic disk drive, magnetic floppy disk or tape, optical disk such as a CD or DVD ROM, or a removable optical drive, etc.

[0048] The client application **105** comprises at least a "wallet" function. This has two main functionalities. One of these is to enable the respective user party **103** to create, sign and send transactions **152** to be propagated throughout the network of nodes **104** and thereby included in the blockchain **150**. The other is to report back to the respective party the amount of the digital asset that he or she currently owns. In an output-based system, this second functionality comprises collating the amounts defined in the outputs of the various **152** transactions scattered throughout the blockchain **150** that belong to the party in question.

[0049] The instance of the client application **105** on each computer equipment **102** is operatively coupled to at least one of the forwarding nodes **104**F of the P2P network **106**. This enables the wallet function of the client **105** to send transactions **152** to the network **106**. The client **105** is also able to contact one, some or all of the storage nodes **104** in order to query the blockchain **150** for any transactions of which the respective party **103** is the recipient (or indeed inspect other parties' transactions in the blockchain **150**, since in embodiments the blockchain **150** is a public facility which provides trust in transactions in part through its public visibility). The wallet function on each computer equipment **102** is configured to formulate and send transactions **152** according to a transaction protocol. Each node **104** runs software configured to validate transactions **152** according to a node protocol, and in the case of the forwarding nodes **104**F to forward transactions **152** in order to propagate them throughout the network **106**. The transaction protocol and node protocol correspond to one another, and a given transaction protocol goes with a given node protocol, together implementing a given transaction model. The same transaction protocol is used for all transactions **152** in the blockchain **150** (though the transaction protocol may allow different subtypes of transaction within it). The same node protocol is used by all the nodes **104** in the network **106** (though it many handle different subtypes of transaction differently in accordance with the rules defined for that subtype, and also different nodes may take on different roles and hence implement different corresponding aspects of the protocol).

[0050] As mentioned, the blockchain **150** comprises a chain of blocks **151**, wherein each block **151** comprises a set of one or more transactions **152** that have been created by a proof-of-work process as discussed previously. Each block **151** also comprises a block pointer **155** pointing back to the previously created block **151** in the chain so as to define a sequential order to the blocks **151**. The blockchain **150** also comprises a pool of valid transactions **154** waiting to be included in a new block by the proof-of-work process. Each transaction **152** (other than a generation transaction) comprises a pointer back to a previous transaction so as to define an order to sequences of transactions (N.B. sequences of transactions **152** are allowed to branch). The chain of blocks **151** goes all the way back to a genesis block (Gb) **153** which was the first block in the chain. One or more original transactions **152** early on in the chain **150** pointed to the genesis block **153** rather than a preceding transaction.

[0051] When a given party **103**, say Alice, wishes to send a new transaction **152*j*** to be included in the blockchain **150**, then she formulates the new transaction in accordance with the relevant transaction protocol (using the wallet function in her client application **105**). She then sends the transaction **152** from the client application **105** to one of the one or more forwarding nodes **104**F to which she is connected. E.g. this could be the forwarding node **104**F that is nearest or best connected to Alice's computer **102**. When any given node **104** receives a new transaction **152*j***, it handles it in accordance with the node protocol and its respective role. This comprises first checking whether the newly received transaction **152*j*** meets a certain condition for being "valid", examples of which will be discussed in more detail shortly. In some transaction protocols, the condition for validation may be configurable on a per-transaction basis by scripts included in the transactions **152**. Alternatively, the condition could simply be a built-in feature of the node protocol, or be defined by a combination of the script and the node protocol.

[0052] On condition that the newly received transaction **152*j*** passes the test for being deemed valid (i.e. on condition that it is "validated"), any storage node **104**S that receives the transaction **152*j*** will add the new validated transaction **152** to the pool **154** in the copy of the blockchain **150** maintained at that node **104**S. Further, any forwarding node **104**F that receives the transaction **152*j*** will propagate the validated transaction **152** onward to one or more other nodes **104** in the P2P network **106**. Since each forwarding node **104**F applies the same protocol, then assuming the transaction **152*j*** is valid, this means it will soon be propagated throughout the whole P2P network **106**.

[0053] Once admitted to the pool **154** in the copy of the blockchain **150** maintained at one or more storage nodes **104**, then miner nodes **104**M will start competing to solve the proof-of-work puzzle on the latest version of the pool **154** including the new transaction **152** (other miners **104**M may still be trying to solve the puzzle based on the old view of the pool **154**, but whoever gets there first will define where the next new block **151** ends and the new pool **154** starts, and eventually someone will solve the puzzle for a part of the pool **154** which includes Alice's transaction **152*j***). Once the proof-of-work has been done for the pool **154** including the new transaction **152*j***, it immutably becomes part of one of the blocks **151** in the blockchain **150**. Each transaction **152** comprises a pointer back to an earlier transaction, so the order of the transactions is also immutably recorded.

[0054] FIG. **2** illustrates an example transaction protocol. This is an example of an UTXO-based protocol. A transaction **152** (abbreviated "Tx") is the fundamental data structure of the blockchain **150** (each block **151** comprising one or more transactions **152**). The following will be described by reference to an output-based or "UTXO" based protocol. However, this not limiting to all possible embodiments.

[0055] In a UTXO-based model, each transaction ("Tx") **152** comprises a data structure comprising one or more inputs **202**, and one or more outputs **203**. Each output **203** may comprise an unspent transaction output (UTXO), which can be used as the source for the input **202** of another new transaction (if the UTXO has not already been redeemed). The UTXO specifies an amount of a digital asset (a store of value). It may also contain the transaction ID of the transaction from which it came, amongst other information. The transaction data structure may also comprise a header **201**, which may comprise an indicator of the size of the input field(s) **202** and output field(s) **203**. The header **201** may also include an ID of the transaction. In embodiments the transaction ID is the hash of the transaction data (excluding the transaction ID itself) and stored in the header **201** of the raw transaction **152** submitted to the miners **104**M.

[0056] Say Alice **103***a* wishes to create a transaction **152***j* transferring an amount of the digital asset in question to Bob **103***b*. In FIG. **2** Alice's new transaction **152***j* is labelled "Tx_1". It takes an amount of the digital asset that is locked to Alice in the output **203** of a preceding transaction **152***i* in the sequence, and transfers at least some of this to Bob. The preceding transaction **152***i* is labelled "T_{X0}" in FIG. **2**. T_{X0} and Tx_1 are just arbitrary labels. They do not necessarily mean that T_{X0} is the first transaction in the blockchain **151**, nor that Tx_1 is the immediate next transaction in the pool **154**. Tx_1 could point back to any preceding (i.e. antecedent) transaction that still has an unspent output **203** locked to Alice.

[0057] The preceding transaction T_{X0} may already have been validated and included in the blockchain **150** at the time when Alice creates her new transaction Tx_1, or at least by the time she sends it to the network **106**. It may already have been included in one of the blocks **151** at that time, or it may be still waiting in the pool **154** in which case it will soon be included in a new block **151**. Alternatively, T_{X0} and Tx_1 could be created and sent to the network **102** together, or T_{X0} could even be sent after Tx_1 if the node protocol allows for buffering "orphan" transactions. The terms "preceding" and "subsequent" as used herein in the context of the sequence of transactions refer to the order of the transactions in the sequence as defined by the transaction pointers specified in the transactions (which transaction points back to which other transaction, and so forth). They could equally be replaced with "predecessor" and "successor", or "antecedent" and "descendant", "parent" and "child", or such like. It does not necessarily imply an order in which they are created, sent to the network **106**, or arrive at any given node **104**. Nevertheless, a subsequent transaction (the descendent transaction or "child") which points to a preceding transaction (the antecedent transaction or "parent") will not be validated until and unless the parent transaction is validated. A child that arrives at a node **104** before its parent is considered an orphan. It may be discarded or buffered for a certain time to wait for the parent, depending on the node protocol and/or miner behaviour.

[0058] One of the one or more outputs **203** of the preceding transaction T_{X0} comprises a particular UTXO, labelled here $UTXO_0$. Each UTXO comprises a value specifying an amount of the digital asset represented by the UTXO, and a locking script which defines a condition which must be met by an unlocking script in the input **202** of a subsequent transaction in order for the subsequent transaction to be validated, and therefore for the UTXO to be successfully redeemed. Typically, the locking script locks the amount to a particular party (the beneficiary of the transaction in which it is included). I.e. the locking script defines an unlocking condition, typically comprising a condition that the unlocking script in the input of the subsequent transaction comprises the cryptographic signature of the party to whom the preceding transaction is locked.

[0059] The locking script (aka scriptPubKey) is a piece of code written in the domain specific language recognized by the node protocol. A particular example of such a language is called "Script" (capital S). The locking script specifies what information is required to spend a transaction output **203**, for example the requirement of Alice's signature. Unlocking scripts appear in the outputs of transactions. The unlocking script (aka scriptSig) is a piece of code written the domain specific language that provides the information required to satisfy the locking script criteria. For example, it may contain Bob's signature. Unlocking scripts appear in the input **202** of transactions.

[0060] So, in the example illustrated, $UTXO_0$ in the output **203** of T_{X0} comprises a locking script [Checksig P_A] which requires a signature Sig P_A of Alice in order for $UTXO_0$ to be redeemed (strictly, in order for a subsequent transaction attempting to redeem $UTXO_0$ to be valid). [Checksig P_A] contains the public key P_A from a public-private key pair of Alice. The input **202** of Tx_1 comprises a pointer pointing back to Tx_1 (e.g. by means of its transaction ID, $TxID_0$, which in embodiments is the hash of the whole transaction T_{X0}). The input **202** of Tx_1 comprises an index identifying $UTXO_0$ within T_{X0}, to identify it amongst any other possible outputs of T_{X0}. The input **202** of Tx_1 further comprises an unlocking script <Sig P_A>which comprises a cryptographic signature of Alice, created by Alice applying her private key from the key pair to a predefined portion of data (sometimes called the "message" in cryptography). What data (or "message") needs to be signed by Alice to provide a valid signature may be defined by the locking script, or by the node protocol, or by a combination of these.

[0061] When the new transaction Tx_1 arrives at a node **104**, the node applies the node protocol. This comprises running the locking script and unlocking script together to check whether the unlocking script meets the condition defined in the locking script (where this condition may comprise one or more criteria). In embodiments this involves concatenating the two scripts:

$$<\text{Sig } P_A><P_A>\|[\text{Checksig } P_A]$$

where "||" represents a concatenation and "< . . . >" means place the data on the stack, and "[. . .]" is a function comprised by the unlocking script (in this example a stack-based language). Equivalently the scripts may be run one after another, with a common stack, rather than concatenating the scripts. Either way, when run together, the scripts use the public key P_A of Alice, as included in the locking script in the output of T_{X0}, to authenticate that the locking script in the input of Tx_1 contains the signature of Alice signing the expected portion of data. The expected portion of data itself (the "message") also needs to be included in T_{X0} order to perform this authentication. In embodiments the signed data comprises the whole of T_{X0} (so a separate element does to need to be included specifying the signed portion of data in the clear, as it is already inherently present).

[0062] The details of authentication by public-private cryptography will be familiar to a person skilled in the art. Basically, if Alice has signed a message by encrypting it with her private key, then given Alice's public key and the message in the clear (the unencrypted message), another entity such as a node **104** is able to authenticate that the encrypted version of the message must have been signed by Alice. Signing typically comprises hashing the message, signing the hash, and tagging this onto the clear version of the message as a signature, thus enabling any holder of the public key to authenticate the signature.

[0063] If the unlocking script in Tx_1 meets the one or more conditions specified in the locking script of T_{X0} (so in the example shown, if Alice's signature is provided in Tx_1 and authenticated), then the node **104** deems Tx_1 valid. If it is a mining node **104**M, this means it will add it to the pool of transactions **154** awaiting proof-of-work. If it is a forwarding node **104**F, it will forward the transaction Tx_1 to one or more other nodes **104** in the network **106**, so that it will be propagated throughout the network. Once Tx_1 has been validated and included in the blockchain **150**, this defines $UTXO_0$ from T_{X0} as spent. Note that Tx_1 can only be valid if it spends an unspent transaction output **203**. If it attempts to spend an output that has already been spent by another transaction **152**, then Tx_1 will be invalid even if all the other conditions are met. Hence the node **104** also needs to check whether the referenced UTXO in the preceding transaction T_{X0} is already spent (has already formed a valid input to another valid transaction). This is one reason why it is important for the blockchain **150** to impose a defined order on the transactions **152**. In practice a given node **104** may maintain a separate database marking which UTXOs **203** in which transactions **152** have been spent, but ultimately what defines whether a UTXO has been spent is whether it has already formed a valid input to another valid transaction in the blockchain **150**.

[0064] Note that in UTXO-based transaction models, a given UTXO needs to be spent as a whole. It cannot "leave behind" a fraction of the amount defined in the UTXO as spent while another fraction is spent. However, the amount from the UTXO can be split between multiple outputs of the next transaction. E.g. the amount defined in $UTXO_0$ in T_{X0} can be split between multiple UTXOs in Tx_1. Hence if Alice does not want to give Bob all of the amount defined in $UTXO_0$, she can use the remainder to give herself change in a second output of Tx_1 or pay another party.

[0065] In practice Alice will also usually need to include a fee for the winning miner, because nowadays the reward of the generation transaction alone is not typically sufficient to motivate mining. If Alice does not include a fee for the miner, T_{X0} will likely be rejected by the miner nodes **104**M, and hence although technically valid, it will still not be propagated and included in the blockchain **150** (the miner protocol does not force miners **104**M to accept transactions **152** if they don't want). In some protocols, the mining fee does not require its own separate output **203** (i.e. does not need a separate UTXO). Instead any different between the total amount pointed to by the input(s) **202** and the total amount of specified in the output(s) **203** of a given transaction **152** is automatically given to the winning miner **104**. E.g. say a pointer to $UTXO_0$ is the only input to Tx_1, and Tx_1 has only one output $UTXO_1$. If the amount of the digital asset specified in $UTXO_0$ is greater than the amount specified in $UTXO_1$, then the difference automatically goes to the winning miner **104**M. Alternatively or additionally however, it is not necessarily excluded that a miner fee could be specified explicitly in its own one of the UTXOs **203** of the transaction **152**.

[0066] Note also that if the total amount specified in all the outputs **203** of a given transaction **152** is greater than the total amount pointed to by all its inputs **202**, this is another basis for invalidity in most transaction models. Therefore, such transactions will not be propagated nor mined into blocks **151**.

[0067] Alice and Bob's digital assets consist of the unspent UTXOs locked to them in any transactions **152** anywhere in the blockchain **150**. Hence typically, the assets of a given party **103** are scattered throughout the UTXOs of various transactions **152** throughout the blockchain **150**. There is no one number stored anywhere in the blockchain **150** that defines the total balance of a given party **103**. It is the role of the wallet function in the client application **105** to collate together the values of all the various UTXOs which are locked to the respective party and have not yet been spent in another onward transaction. It can do this by querying the copy of the blockchain **150** as stored at any of the storage nodes **104**S, e.g. the storage node **104**S that is closest or best connected to the respective party's computer equipment **102**.

[0068] Note that the script code is often represented schematically (i.e. not the exact language). For example, one may write [Checksig P_A] to mean [Checksig P_A]=OP_DUP OP_HASH160<H(Pa)>OP_EQUALVERIFY OP_CHECKSIG. "OP_. . . " refers to a particular opcode of the Script language. OP_CHECKSIG (also called "Checksig") is a Script opcode that takes two inputs (signature and public key) and verifies the signature's validity using the Elliptic Curve Digital Signature Algorithm (ECDSA). At runtime, any occurrences of signature ('sig') are removed from the script but additional requirements, such as a hash puzzle, remain in the transaction verified by the 'sig' input. As another example, OP_RETURN is an opcode of the Script language for creating an unspendable output of a transaction that can store metadata within the transaction, and thereby record the metadata immutably in the blockchain **150**. E.g. the metadata could comprise a document which it is desired to store in the blockchain.

[0069] The signature P_A is a digital signature. In embodiments this is based on the ECDSA using the elliptic curve secp**256***k***1**. A digital signature signs a particular piece of data. In embodiments, for a given transaction the signature will sign part of the transaction input, and all or part of the transaction output. The particular parts of the outputs it signs depends on the SIGHASH flag. The SIGHASH flag is a 4-byte code included at the end of a signature to select which outputs are signed (and thus fixed at the time of signing).

[0070] The locking script is sometimes called "scriptPubKey" referring to the fact that it comprises the public key of the party to whom the respective transaction is locked. The unlocking script is sometimes called "scriptSig" referring to the fact that it supplies the corresponding signature. However, more generally it is not essential in all applications of a blockchain **150** that the condition for a UTXO to be redeemed comprises authenticating a signature. More generally the scripting language could be used to define any one or more conditions. Hence the more general terms "locking script" and "unlocking script" may be preferred.

Optional Side Channel

[0071] FIG. **3** shows a further system **100** for implementing a blockchain **150**. The system **100** is substantially the same as that described in relation to FIG. **1** except that additional communication functionality is involved. The client application on each of Alice and Bob's computer equipment **102***a*, **120***b*, respectively, comprises additional communication functionality. That is, it enables Alice **103***a* to establish a separate side channel **301** with Bob **103***b* (at the instigation of either party or a third party). The side channel **301** enables exchange of data separately from the P2P network. Such communication is sometimes referred to as "off-chain". For instance, this may be used to exchange a transaction **152** between Alice and Bob without the transaction (yet) being published onto the network P2P **106** or making its way onto the chain **150**, until one of the parties chooses to broadcast it to the network **106**. Alternatively, or additionally, the side channel **301** may be used to exchange any other transaction related data, such as keys, negotiated amounts or terms, data content, etc.

[0072] The side channel **301** may be established via the same packet-switched network **101** as the P2P overlay network **106**. Alternatively, or additionally, the side channel **301** may be established via a different network such as a mobile cellular network, or a local area network such as a local wireless network, or even a direct wired or wireless link between Alice and Bob's devices **1021**, **102***b*. Generally, the side channel **301** as referred to anywhere herein may comprise any one or more links via one or more networking technologies or communication media for exchanging data "off-chain", i.e. separately from the P2P overlay network **106**. Where more than one link is used, then the bundle or collection of off-chain links as a whole may be referred to as the side channel **301**. Note therefore that if it is said that Alice and Bob exchange certain pieces of information or data, or such like, over the side channel **301**, then this does not necessarily imply all these pieces of data have to be send over exactly the same link or even the same type of network.

Node Software

[0073] FIG. **4** illustrates an example of the node software **400** that is run on each node **104** of the P2P network **106**, in the example of a UTXO- or output-based model. The node software **400** comprises a protocol engine **401**, a script engine **402**, a stack **403**, an application-level decision engine **404**, and a set of one or more blockchain-related functional modules **405**. At any given node **104**, these may include any one, two or all three of: a mining module **405**M, a forwarding module **405**F and a storing module **405**S (depending on the role or roles of the node). The protocol engine **401** is configured to recognize the different fields of a transaction **152** and process them in accordance with the node protocol. When a transaction **152***m* (Tx_m) is received having an input pointing to an output (e.g. UTXO) of another, preceding transaction **152***m***-1** (Tx_{m-1}), then the protocol engine **401** identifies the unlocking script in Tx_m and passes it to the script engine **402**. The protocol engine **401** also identifies and retrieves Tx_{m-1} based on the pointer in the input of Tx_m. It may retrieve Tx_{m-1} from the respective node's own pool **154** of pending transactions if Tx_{m-1} is not already on the blockchain **150**, or from a copy of a block **151** in the blockchain **150** stored at the respective node or another node **104** if Tx_{m-1} is already on the blockchain **150**. Either way, the script engine **401** identifies the locking script in the pointed-to output of Tx_{m-1} and passes this to the script engine **402**.

[0074] The script engine **402** thus has the locking script of Tx_{m-1} and the unlocking script from the corresponding input of Tx_m. For example, Tx_1 and Tx_2 are illustrated in FIG. **4**, but the same could apply for any pair of transactions, such as Tx, and Tx_1, etc. The script engine **402** runs the two scripts together as discussed previously, which will include placing data onto and retrieving data from the stack **403** in accordance with the stack-based scripting language being used (e.g. Script).

[0075] By running the scripts together, the script engine **402** determines whether the unlocking script meets the one or more criteria defined in the locking script—i.e. does it "unlock" the output in which the locking script is included? The script engine **402** returns a result of this determination to the protocol engine **401**. If the script engine **402** determines that the unlocking script does meet the one or more criteria specified in the corresponding locking script, then it returns the result "true". Otherwise it returns the result "false".

[0076] In an output-based model, the result "true" from the script engine **402** is one of the conditions for validity of the transaction. Typically there are also one or more further, protocol-level conditions evaluated by the protocol engine **401** that must be met as well; such as that the total amount of digital asset specified in the output(s) of Tx_m does not exceed the total amount pointed to by the input(s), and that the pointed-to output of Tx_{m-1} has not already been spent by another valid transaction. The protocol engine **401** evaluates the result from the script engine **402** together with the one or more protocol-level conditions, and only if they are all true does it validate the transaction Tx_m. The protocol engine **401** outputs an indication of whether the transaction is valid to the application-level decision engine **404**. Only on condition that Tx_m is indeed validated, the decision engine **404** may select to control one or both of the mining module **405**M and the forwarding module **405**F to perform their respective blockchain-related function in respect of Tx_m. This may comprise the mining module **405**M adding Tx_m to the node's respective pool **154** for mining into a block **151**, and/or the forwarding module **405**F forwarding Tx_m to another node **104** in the P2P network **106**. Note however that in embodiments, while the decision engine **404** will not select to forward or mine an invalid transaction, this does not necessarily mean that, conversely, it is obliged to trigger the mining or the forwarding of a valid transaction simply because it is valid. Optionally, in embodiments the decision engine **404** may apply one or more additional conditions before triggering either or both functions. E.g. if the node is a mining node **104**M, the decision engine may only select to mine the transaction on condition that the transaction is both valid and leaves enough of a mining fee.

[0077] Note also that the terms "true" and "false" herein do not necessarily limit to returning a result represented in the form of only a single binary digit (bit), though that is certainly one possible implementation. More generally, "true" can refer to any state indicative of a successful or affirmative outcome, and "false" can refer to any state indicative of an unsuccessful or non-affirmative outcome.

For instance in an account-based model (not illustrated in FIG. **4**), a result of "true" could be indicated by a combination of an implicit, protocol-level) validation of a signature by the node **104** and an additional affirmative output of a smart contract (the overall result being deemed to signal true if both individual outcomes are true).

Script

[0078] Blockchain protocols may use a scripting language for transactions. A script is essentially a list of elements, which may be data or instructions. The instructions are referred to in the literature as, script words, opcodes, commands, or functions. Opcodes (short for operation codes) perform predefined operations on the data within a script. The data within a script may be, for example, numbers, public keys, signatures, hash values, etc. An opcode is a function that operates on the data within a script. In scripting language, a script is run from one end to the other (usually from left to right) and makes use of a data structure referred to as a "stack". Data is always pushed to (i.e. placed on) the stack. An opcode can pop data off the stack (i.e. take data from the stack), perform an operation on the data, and then optionally "push" new data on to the stack. The stack-based scripting language commonly used in a number of blockchains is just called Script. The following will be described in terms of opcodes of the Script language.

[0079] Stack-based scripting languages will be familiar to the person skilled in the art. The following example illustrates an example script implementation. Specifically, an example verification and unlocking process is shown.

[0080] An example script may comprise <Bob's signature><Bob's public key>OP_DUP OP_HASH <Bob's public address>OP_EQUALVERIFY OP_CHECKSIG. The script is operated on from left to right.

[0081] Step 1: Push <Bob's signature>on to the stack

<Bob's signature>

[0082] Step 2: Push <Bob's public key>on to the stack (this is now the top element on the stack)

<Bob's public key>
<Bob's signature>

[0083] Step 3: The OP_DUP opcode operates on the top element on the stack to duplicate <Bob's public key>.

<Bob's public key>
<Bob's public key>
<Bob's signature>

[0084] Step 4: The OP_HASH opcode pops out <Bob's public key> and runs it through a hash algorithm (followed by one or more optional operations) to get <Bob's public address> and place it on the stack.

<Bob's public address>
<Bob's public key>
<Bob's signature>

[0085] Step 5: Push <Bob's public address> to the stack (this is now the top element on the stack).

<Bob's public address>
<Bob's public address>
<Bob's public key>
<Bob's signature>

[0086] Step 6: The OP_EQUALVERIFY opcode pops the last two elements off the stack (<Bob's public address> and <Bob's public address>) and checks to see if the two addresses are identical or not. If they are not identical the execution is considered as failed. If the condition is TRUE, the next command gets executed.

<Bob's public key>
<Bob's signature>

[0087] Step 7: The OP_CHECKSIG opcode pops out <Bob's public key> and <Bob's signature> and checks to see their validity. When this process is complete, Bob can unlock the transaction and access the specified amount of digital asset.

TRUE

[0088] For brevity, the opcodes used herein will be ones of the Script language, e.g. "OP_CHECKSIG". However, the disclosure is not limited to opcodes of the Script language. More generally, while embodiments will be described in terms of opcodes of a specific blockchain scripting language (Script), the same teaching can be implemented using any opcode of any scripting language which when called by a script engine (e.g. a script interpreter) performs a particular function. Also while embodiments below may be exemplified in terms of an elliptic curve digital signature algorithm (ECDSA), it will be appreciated that other cryptographic signature schemes such as RSA are known in the art and could equally be employed herein.

Binary Threshold Protocol

[0089] An example of a threshold based unlocking system is that of the multi-signature functionality of some blockchain protocols. Within some blockchain protocols, it is possible to construct a transaction locking script that can be unlocked by providing any m-of-n ECDSA signatures corresponding to m previously specified public keys. The locking script condition for such a multi-signature transaction is written as

$m <P_1> <P_2> \ldots <P_{n-1}> <P_n>$ n OP_CHECKMULTISIG

[0090] Here the locking scripts begins with the number of valid signatures that are required (m), then lists the set of n public keys for which m signatures must correspond, then the value n, and the opcode OP_CHECKMULTISIG that checks if all m signatures are valid.

[0091] The unlocking script would be of the following form.

0 <sigP_a>...<sigP_m>

[0092] This includes the m signatures as well as a placeholder value **0**. The placeholder is not processed per se, it is only required due to an existing bug in a current implementation of a particular blockchain protocol.

[0093] Embodiments of the present disclosure provide techniques for constructing a transaction locking script (referred to synonymously as an output script or scriptPubKey) that can be unlocked by providing, in an unlocking script (referred to synonymously as an input script of scriptSig) of a later transaction, data items that satisfy any m-of-n criteria. Like

[0094] OP_CHECKMULTISIG, the criteria may require signatures corresponding to public keys. However, the following techniques are not limited to requiring signatures. In general, any criteria may be defined. For instance, there may be a requirement to provide a pre-image to a hash, a solution to an equation, an answer to a quiz question, and so on.

[0095] Embodiments will first be described with reference to FIG. **5**. FIG. **5** is a simplified version of FIG. **1** and illustrates an example system **500** for implementing a Binary Threshold Protocol (BTP). A first party (Alice) **103***a* generates a transaction Tx_1 and transmits the first transaction Tx_1 to the blockchain network **106**. Assuming the blockchain protocol is satisfied, the first transaction Tx_1 will be included in the blockchain **150**. Note that instead, a different party could transmit Alice's transaction to the blockchain network **106** on her behalf. The first transaction Tx_1 comprises an output which specifies an amount of a digital asset. The digital asset is locked by an output script (or locking script). In order to be unlocked and transferred to a second party (Bob) **103***b*, Bob must transmit a second transaction Tx_2 to the blockchain network **106** which has an input script (or unlocking script) which can unlock the output script of the first transaction Tx_1.

[0096] When constructing the output script, Alice **103***a* includes a plurality of criterion components. Each criterion component comprises one or more opcodes (or functions in general), e.g. OP_CHECKSIGVERIFY. A criterion component may also comprise one or more data elements, e.g. a hash value. Each criterion component defines a criterion which requires a specific input data item to be satisfied. That is, each criterion component, when run against an input script of a later transaction, requires that input script to include a specific data item in order for the criterion component to successfully execute, i.e. return a value representing "TRUE", e.g. <1>. For example, some or all of the criterion components may require a different digital signature to be provided as an input data item. Preferably, each criterion component may define a different criterion. Alternatively, one or more of the criterion components may define the same criterion.

[0097] Alice **103***a* also includes, in the output script, a plurality of counter script components. Each counter script component is associated with a respective one of the criterion components. For instance, each counter script component may, in the case of a scripting language where the script is executed from left to right, follow a respective one of the criterion components. Each counter components comprises one or more opcodes (or functions) and, optionally, one or more data items. Each counter component is configured to increment a counter if its associated criterion component has been satisfied. That is, if a first criterion component is satisfied, a first counter component will increment the counter. Note here that "increment" may be taken to be mean increase or decrease. For instance, the counter may be incremented in an increasing manner (e.g. one, two, three, etc.) or the counter may be incremented in a decreasing manner (e.g. three, two, one, etc.). In some examples, each counter component may increment the counter by the same amount, i.e. by the same number. For instance, the counter may increase by one each time a criterion component is satisfied. Alternatively, the counter components may be weighted. That is, some counter components may increment the counter by different amounts, i.e. by different numbers. As an example, a first counter component may increment the counter by one, whilst a second counter component may increment the counter by two.

[0098] Alice **103***a* constructs the output script such that the counter must reach at least a predetermined number (or threshold) for the output to be unlocked. That is, the output script is configured to determine whether the counter has incremented to a predetermined number. If so, the output may be unlocked and redeemed by Bob **103***b*. If the threshold is not reached, the output will not be unlocked. In the case that the counter is incremented in an increasing manner, the output script is configured to determine whether the counter is equal to or greater than the threshold. Conversely, in the case that the counter is incremented in a decreasing manner, the output script is configured to determine whether the counter is less than or equal to the threshold. The counter script may comprise one or more opcodes and optionally one or more data elements which together perform a check on the counter and successfully evaluate as "TRUE" if the threshold is reached.

[0099] In some examples, the total number of criterion components may be more than the threshold number. In other words, it may be possible to increment the counter, by satisfying criterion components, to a number greater than the predetermined number necessary for unlocking the output. Alternatively, each criterion component must be satisfied in order for the output to be unlocked.

[0100] Optionally, a given criterion component may be made up of two or more sub-criterion components. Like criterion components, sub-criterion components comprise opcodes and optionally data elements, which together require an input data item of a later transaction to successfully evaluate. As an example, a criterion component may be made up of multiple sub-criterion components, each of which apply a hash function to a respective input data item and compare the resulting hash values to predetermined hash values. If the resulting and predetermined hash values match, the sub-criterion components and therefore the criterion component are satisfied. Note that even if a criterion component comprises multiple sub-components, the counter is incremented only once when the criterion component as a whole is satisfied, not once for each sub-criterion component.

[0101] As shown in FIG. **5**, Alice **103***a* may transmit the criteria Q_n to Bob **103***b*. Note that the criteria On define what is required as input data items to satisfy the criterion components, but not the data items themselves. For instance, a criterion component may require a digital signature, but would not include the digital signature itself. By providing Bob **103***b* with the criteria Q_n, Bob can know what data items he needs to provide. Alternatively, Alice **103***a* may not provide the criteria to Bob **103***b*, in which case Bob would need to interpret the criterion components (e.g. a sequence of opcodes) to determine what should be provided as a corresponding data item.

[0102] The output script may comprise n criterion components. If the recipient (Bob) satisfies at least m of the n criteria formalized in the locking script he can redeem the full amount of the digital asset being offered by Alice **103***a*. If he does not satisfy at least m criteria Bob **103***b* cannot redeem any of the digital asset. The BTP represents an m-of-n criteria threshold where m is fixed and the recipient can only access the digital asset if they satisfy at least m criteria. The protocol is elaborated on below with respect to two individuals Alice **103***a* and Bob **103***b* where Alice **103***a* pays Bob **103***b* if Bob can satisfy at least m of n criteria. Alice **103***a* creates a 'binary-pay' transaction (titled bPay) within which exists an output containing an output script that locks the digital asset contained in the output until Bob **103***b* is able to provide an input script of a spending transaction that includes content that satisfies at least m of the n criteria (in some examples, the input script must also comprise Bob's digital signature). The content of Bob's input script would be a composite of multiple sub-units of input data items. Each data item is offered as a solution for a particular criterion.

[0103] This protocol achieves its objectives via an output script in the bPay transaction where, given that an input (from Bob) satisfies its corresponding criteria in the bPay output script, it increments a counter (e.g. initialised at zero and incremented by one). The criteria are described as 'corresponding' due to the fact that the intended recipient (Bob) is expected to place the input of interest in a location in the input transaction so that that input is compared against the appropriate criteria in the bPay output script of the bPay transaction. The value of the counter may be stored on a stack, e.g. an ALT stack of the Script language. When an input is successful with respect to its criterion then the current state of the counter is removed (popped) from the ALT stack, placed (pushed) on the main stack, incremented, then popped from the main stack and pushed back unto the alt stack. If the final value of the counter is at least m then the recipient Bob **103***b* can unlock the bPay output.

[0104] An example of the output and input scripts that would satisfy the desired characteristics of the BTP are shown below in $TxID_{bPay}$ and $TxID_{spend}$ respectively.

[0105] The subsection [Criterion_1] of the locking (output) script represents a criterion (i.e. condition) that the recipient Bob **103***b* is being asked to satisfy. This could, for instance, ask for the preimage of a hash as shown below.

[Criterion_1] = OP_HASH160 H(PreImage) OP_EQUAL

[0106] The criterion could also represent a 'grouped set of sub-criteria' that, when combined, must evaluate to true.

[Criterion_i] = f([Criterion_i_a] Criterion_i_b], ... , Criterion_i_z])

[0107] Even if [Criterion_i] is a composition of multiple sub-criteria, if the combination of all these sub-criteria evaluate to true, the counter is still only incremented by one unit.

$TxID_{bPay}$	
Inputs	Outputs
<sigA> <PubA>	<0> OP_TOALTSTACK (default counter to 0) [Criterion_1] (criteria 1 subscript) OP_IF OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_ENDIF (if criteria satisfied pop stack, increment counter, return to stack) [Criterion_2] (criteria 2 subscript) OP_IF OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_ENDIF ... [Criterion_n] OP_IF OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_ENDIF OP_FROMALTSTACK <M> OP_GREATERTHANOREQUAL (if counter ≥ m) OP_DUP OP_HASH160 <H(PubB)> OP_EQUALVERIFY OP_CHECKSIG (check Bob's signature)

[0108] Note that the text in parentheses are comments and would not actually be included in the transaction.

[0109] To unlock the output of bPay the recipient Bob **103***b* would provide, in the input script of the spending transaction, n pieces of input where at least m of these inputs satisfy at least m of the previously stated criteria. An example spending transaction $TxID_{bPay}$ is shown above.

$TxID_{Spend}$	
Inputs	Outputs
<sigB> <PubB> <input_n> ... <input_2> <input_1>	

[0110] These set of n inputs are placed in the necessary order such that each of the pieces of input is matched and assessed against the intended criteria when the input and output scripts are combined and executed in the stack. e.g.

[criterion_2] <input_2>

[0111] Similar to criteria subscripts, an<input_i>could also represent a grouped set of inputs.

<input_i> = <input_i_a> <input_i_b> ... <input_i_z>>

[0112] Given the user's ability to define any criterion, this accomplishes the objectives of the binary threshold protocol, where a threshold of generic criteria must be satisfied in order for an output of a transaction to be spent.

Spectrum Threshold Protocol

[0113] The BTP sets a binary threshold for redeeming an amount of a digital asset. The spectrum threshold protocol (STP) sets a range of thresholds, where the amount of the digital asset that can be received by a recipient depends on number of successfully met criteria.

[0114] The example systems of FIGS. **1** and **5** may be used to implement the techniques described below. A first party (Alice) **103***a* constructs a first transaction (referred to as an "sPay transaction") having multiple outputs. Alice **103***a* constructs each output in a similar manner to the BTP. That is, each output has an output script and each output script includes multiple criterion components and multiple counter components, each associated with a respective one of the criterion components. Each criterion component defines a criterion which can be satisfied with an input data item from an input script of a later transaction. Each counter component is configured to increment a counter if its associated criterion component is satisfied. Each counter component may increment the counter by the same amount (e.g. one). Alternatively, some of the counter components may increment the counter by different amounts. The multiple criterion components and the multiple counter components are defined in a first script condition of each output script. Here, the first script condition is merely a label for the part of the output script comprising the criterion components and the counter components. Note that each output comprises the same criterion components.

[0115] Like the BTP, for the STP Alice **103***a* constructs each output script such that the counter must reach at least a predetermined number (or threshold) for the corresponding output to be unlocked. That is, each output script is configured to determine whether the counter has incremented to a predetermined number. If so, the associated output may be unlocked and redeemed by Bob **103***b*. If the threshold is not reached, the output will not be unlocked. Unlike the BTP, at least one output script for the STP requires the counter to reach a different predetermined number. That is, some of the output scripts require different thresholds to be met in order to be unlocked.

[0116] In some examples, each output script requires a different threshold to be met. In other examples, one or more output scripts may share the same threshold. In some examples, each output may lock a different amount of the digital asset. In other examples, one or more output scripts may lock the same amount of the digital asset. The amount of digital asset may increase with the threshold, i.e. a larger amount of digital asset can be redeemed if more criteria can be satisfied. The increase may or may not be proportional to the number of criteria satisfied.

[0117] Like the BTP, for the STP the total number of criterion components may be more than the threshold number. Alternatively, each criterion component must be satisfied in order for the output to be unlocked. Also similar to the BTP, for the STP one or more of the criterion components may be made up of multiple sub-criterion components.

[0118] FIG. **6** illustrates an example representation of the first transaction Tx_1. The first transaction Tx_1 has multiple outputs i. In the example shown, each output locks an amount of digital asset y_i. That is output **1** locks an amount y_1, output **2** locks an amount y_2, and output I locks an amount yr. As shown, the transaction Tx_1 comprises an input comprising an amount of the digital asset which is at least equal to the sum of y_i.

[0119] Note that FIG. **6** illustrates what must be provided in an input script to unlock each output, rather than what it is contained in the output itself. Looking at output **1** of FIG. **6**, to unlock output **1**, an input script of a second transaction Tx_2 may be provided which comprises enough input data items (represented by <input>) to satisfy the threshold for output **1**. The second transaction Tx_2 is referred to below as a "Collect transaction", as it allows Bob **103***b* to collect the digital asset y_1 locked by output **1**. Each first script condition (referred to below as a "Collect condition" may also require the input script of the Collect transaction to comprise Alice and/or Bob's digital signatures, σ_{Alice} and σ_{Bob} respectively. As another optional feature, the Collect condition may require the Collect transaction Tx_2 to comprise a secret value k of Bob's choosing. The secret value is secret in the sense that until the Collect transaction is transmitted to and included in the blockchain **150**, the secret value is known only to Bob **103***b*. For the Collect transaction to require the secret value, Bob **103***b* must transmit a hash of the secret value to Alice **103***a*. The Collect condition can then hash the secret value and compare the resulting value with the hash, obtained from Bob **103***b*, in the Collect condition. Note that there are other ways of unlocking an output of Tx_1. These will be described in detail below with reference to "Refund transactions" and "Cancel transactions".

[0120] Although for convenience the Collect, Refund and Cancel transactions may be written here with capital letters, this does not necessarily imply any particular proprietary or proper-name, and the terms could equally be written with lower-case letters anywhere herein.

[0121] In some examples Alice **103***a* herself may construct part of the Collect transaction Tx_2. For instance, Alice **103***a* may provide, to Bob **103***b* via a side channel (e.g. the side channel **301** shown in FIG. **3**), a partial transaction comprising her digital signature σ_{Alice}. Alice **103***a* may construct such a partial transaction for each of the outputs in the first transaction Tx_1. Alice **103***a* may include a lock time in the partial Collect transaction(s) which she constructs and sends to Bob **103***b* via the side channel. The lock time prevents the Collect transaction from being successfully included in a block of the blockchain **150** until after a certain time (which may e.g. be specified by a Unix time or a block height). The lock time may be implemented using an "nLocktime" field of the transaction. nLocktime is a parameter of a transaction that mandates a minimal time before which the transaction cannot be accepted into a block. Each Collect transaction may specify a different lock time, as discussed below.

[0122] When constructing the first transaction Tx_1, Alice **103***a* may include a second script condition in each output script. The second script condition will be referred to below as a "Refund condition". Like the Collect condition, the Refund condition is a label for a part of the output script and comprises one or more opcodes (or functions in general). The Refund condition may also comprise one or more data elements. The Refund condition is configured to require an input script of a later transaction to comprise Alice's digital signature and Bob's digital signature. That is, the Refund condition determines whether Alice's and Bob's digital signatures have been provided in an input script of a "Refund transaction", and if so, the Refund condition evaluates to TRUE (or <1>), e.g. on the stack.

[0123] Each output is constructed such that an input script of a later transaction can satisfy one script condition only. That is, any given input script cannot satisfy both a Collect condition and a Refund condition. This is because the Collect and Refund conditions require different input data items to be unlocked. For example, the Collect condition requires input data items which satisfy the criterion components whereas the Refund condition does not.

[0124] In some examples Alice **103***a* herself may construct part of the Refund transaction. For instance, Alice **103***a* may provide, to Bob **103***b* via a side channel (e.g. the side channel **301** shown in FIG. **3**), a partial transaction comprising her digital signature σ_{Alice}. Alice **103***a* may construct such a partial transaction for each of the outputs in the first transaction. Alice **103***a* may include a lock time in the partial Refund transaction(s) which she constructs and sends to Bob **103***b* via the side channel. Bob **103***b* may then include his signature O Bob in the partial Refund transaction and return the signed Refund transaction to Alice **103***a*. It will be appreciated that in alternative examples, Bob **103***b* may first sign a partial Refund transaction and send it to Alice for Alice to sign.

[0125] Like the Collect transactions, Alice **103***a* may set a lock time for each of the Refund transactions. The lock time may be set before Alice and/or Bob sign the transaction with their respective signatures. For each output having associated Collect and Refund transactions, the lock time of the Refund transaction will be later than the lock time of the Collect transaction, meaning that the Collect transaction can be included in a block of the blockchain **150** before the associated Refund transaction.

[0126] The Refund condition and Refund transactions together are intended to allow Alice **103***a* to redeem the amount of the digital asset in each output not claimed by Bob **103***b*. As an example, if Bob does not manage to provide an input satisfying the criteria threshold of any output (e.g. within a predetermined period of time), Alice **103***a* needs to be able to reclaim (or Refund) the digital asset tied to the outputs.

[0127] When constructing the first transaction Tx_1 Alice **103***a* may include a third script condition in some or all of the output scripts. The third script condition will be referred to below as a "Cancel condition". Like the Collect and Refund conditions, the Cancel condition is a label for a part of the output script and comprises one or more opcodes (or functions in general). The Cancel condition may also comprise one or more data elements. The Cancel condition is configured to require an input script of a later transaction to comprise Alice's digital signature. That is, the Cancel condition determines whether Alice's digital signature has been provided in an input script of a "Cancel transaction", and if so, the Cancel condition evaluates to TRUE (or <1>), e.g. on the stack. The Cancel condition may also require the secret value k.

[0128] The Cancel condition and Cancel transactions together are intended to allow Alice **103***a* to redeem the amount of the digital asset in each output not claimed by Bob **103***b*. As an example, if one output required two criteria to be satisfied and another output required three criteria to be satisfied, and Bob **103***b* could only satisfy two criteria, Alice **103***a* needs to be able to reclaim (or Refund) the output having an output script that specifies the three-criteria threshold. By requiring a Cancel transaction to comprise the secret value (known only to Bob **103***b* until revealed on the blockchain), Bob **103***b* can be sure that Alice **103***a* cannot cancel any of the transactions until Bob attempts to claim one of the outputs using a Collect transaction.

[0129] A spectrum threshold, in the context of the STP, is utilised where the recipient Bob **103***b* gets paid based on the number of criteria that he is able to satisfy. As per the binary threshold, it does not matter 'which criteria' is satisfied, what matters is the number of criteria that are satisfied.

[0130] To achieve the spectrum threshold functionality, the STP utilises a transaction (referred to as an sPay transaction) which includes at least $\pi \leq n$ outputs. Each output i pays a certain amount y_i based on the amount of criteria, m_i, that Bob is asked to satisfy. The set of m_i values are defined as

$$\text{set}=\{m_i: i \in \{1,2, \ldots, \pi\}, m_i \in \mathbb{N}\}$$

[0131] The intention is that if a recipient is able to satisfy m; criteria, the recipient is able to collect the y; payment from a dedicated output, output i, of the sPay transaction. While the protocol is suitable for such instances, the payment amount does not necessarily have to be proportional to the number of criteria satisfied. As an example of non-proportionality, if Bob **103***b* satisfies at least one criterion Bob can redeem 10 units of the digital asset; if at least two criteria are satisfied then 15 units of the digital asset; and if at least three criteria then 25 units of the digital asset.

[0132] Furthermore, the threshold criteria payments for the set of outputs can be 'arbitrary', and thus do not necessarily have to be an arithmetic progression and/or follow a defined pattern. Consider an sPay transaction's inclusion of outputs for 'at least m_a criteria', for 'at least m_b criteria', for 'at least m_c criteria', and for 'at least m_a criteria' where $m_a<m_p$ and $m_c<m_d$; and no outputs for threshold values m_x, m_y, m_2 exist in the sPay transaction where

$$m_a<m_x<m_b$$

$$m_b<m_y<m_c$$

$$m_c<m_z<m_d$$

[0133] 'No arithmetic progression' would mean that y_b-y_a does not necessarily have to be equal to y_d-y_c. In light of the accommodation of 'arbitrary threshold' criteria for the outputs, recall that although there are n criteria that may be satisfied, there are $\pi \leq n$ outputs.

[0134] In some examples, each output of the sPay transaction comprises multiple script conditions, e.g. Collect, Refund and Cancel conditions. Each output must include the Collect condition and can optionally include the Refund condition and/or the Cancel conditions. As shown in FIG. **6**, output **1** comprises a Collect and Refund condition as it can be satisfied by either a Collect or Refund transaction. Output **2** comprises Collect, Refund and Cancel conditions as it can be satisfied by a Collect, Refund, or Cancel transaction.

[0135] The different spending transactions and a description of each transaction is provided below.

[0136] Collect Transaction-The Collect transaction ($Collect_i$) is the transaction that pays the winnings (y_i) to the recipient (Bob) if Bob satisfies the required criteria threshold for that output i. The input script of the Collect transaction may require one, some or all of: the signature of the payer (Alice), the signature of the recipient (Bob), and a secret value k. This secret value is originally known only to Bob and is only revealed when Bob spends the Collect transaction. The Collect transaction may be time-locked, e.g. the nLockTime field/parameter is assigned a value t_i that prevents the Collect transaction from being successfully submitted to the blockchain **150** before that point in time.

[0137] Cancel Transaction-The Cancel transaction ($Cancel_i$) is the transaction that returns the winnings (y_i) of output i to the payer (Alice) if Bob has previously submitted a Collect transaction to any another output j where j≠i. The input script of the Cancel transaction requires the signature of the payer (Alice) and, optionally, the secret value k. This secret may be retrieved by Alice from the Collect transaction that Bob had previously submitted to the blockchain **150**. This Cancel transaction can be submitted at any point in time by Alice as long as Alice knows k.

[0138] Refund Transaction-The Refund transaction ($Refund_i$) is the transaction that returns the winnings (y_i) of output i to the payer (Alice) if the recipient (Bob) does not submit to the blockchain **150** a Collect transaction (for that output), e.g. before a specified point in time $t_{i,i+1}$, where $t_i < t_{i,i+1} < t_{i+1}$. The input script of the Refund transaction requires the signature of the payer (Alice) and the signature of the recipient (Bob). The Refund transaction may be time-locked, i.e. that the Refund transaction can only be successfully submitted to the blockchain **150** after the nLockTime field/parameter value $t_{i,i+1}$.

[0139] As stated above, the design of the sPay transaction is shown in FIG. **6**. The information represented in the figure is focused on the various outputs of the transaction and what it would require to successfully spend each output. In this example, each output i>1 is locked by an output script that allows the recipient to spend the output using three different options. Each of these unlocking options corresponds to a transaction (Refund, Collect, or Cancel). Note, as previously mentioned, that output **1** can be spent using only two options (Refund and Collect). In addition, while not mandatory, the STP is built on the premise or expectation that the more criteria a recipient is able to satisfy, the more the recipient should get paid. In line with this, for the sPay transaction represented in FIG. **6** outputs are ordered (top to bottom) with respect to the amount being paid out (which coincides with the number of criteria to be satisfied for that output.) In the figure the larger output (and therefore number of criteria required) is above an output with lower a lower output.

[0140] FIG. **7** illustrates an example output of an sPay transaction for an output i: i≠1. This output can be spent using either the Cancel, Collect, or Refund transactions.

[0141] FIG. **8** illustrates an example Cancel transaction **800**. In order to spend the output shown in FIG. **7** utilising the Cancel transaction **800**, the payer (expected to be the payer (Alice), must produce her signature as well as the secret value k. The secret value is conceived and originally only known to the recipient Bob **103***b*. Blockchain scripting languages facilitate the request for an (ECDSA) signature as well the request for the secret value. A hash puzzle may be utilised to request the secret value k. An example of such a locking script that requires both secret value and signature would be

```
OP_HASH160 <H(k)> OP_EQUALVERIFY OP_DUP OP_HASH160
<H(PubA)>
OP_EQUALVERIFY OP_CHECKSIG
```

[0142] where H (k) is the hash of the secret value k and PubA is a public key owned by payer Alice **103***a*. A hash is taken to mean the digest of a one-way cryptographic hashing function such as SHA-256. A typical hash function takes an input of arbitrary size and produces an integer in a fixed range. For example, the SHA-256 hash function gives a 256-bit number as its output hash digest.

[0143] This script is referred to as a Cancel condition ([Condition: Can_i]). The unlocking script (in the Cancel transaction) would be

```
<sigA> <PubA> <k>
```

[0144] where sigA is the (ECDSA) signature of the Cancel transaction utilising the public key PubA. This script is referred to as a Cancel input (<Input: Can_i>).

[0145] FIG. **9** illustrates an example Refund transaction **900**. In order to spend the output shown in FIG. **7** utilising the Refund transaction **900**, the payer (Alice) must produce her signature sigA and the recipient Bob **103***b* must also provide his signature sigB. The Refund transaction **900** must be signed by both payer and recipient; this is to ensure a mutual agreement that the Refund transaction is constructed as it should, with particular attention paid to the required inclusion of an nLockTime value (i.e. a lock time **901**) that prevents the Refund transaction **900** from being valid until after an agreed-upon point in time. An example of a script that would require both signatures would be

```
OP_DUP OP_HASH160 <H(PubA)> OP_EQUALVERIFY
OP_CHECKSIG OP_DUP
OP_HASH160 <H(PubB)> OP_EQUALVERIFY OP_CHECKSIG
```

[0146] This script is referred to as a Refund condition ([Condition: Ref*i*]). A 2-of-2 (m-of-n) multisig script could be utilised instead.

[0147] The unlocking script (in the Refund transaction **900**) would be

```
<sigB> <PubB> <sigA> <PubA>
```

[0148] This script is referred to as a Refund input (<Input: Ref_i>).

[0149] FIG. **10** illustrates an example of a Collect transaction **1000**. in order to spend the output of FIG. **7** utilising the Collect transaction **1000** at least four items must be provided. The payer (Alice) must produce her signature (sigA) whereas the recipient Bob **103***b* must provide his signature (sigB) as well as the secret value k. In addition, Bob **103***b* must provide the inputs,

```
<answers> ≡ <input_n> ... <input_2> <input_1>
```

[0150] from which there are expected to be enough 'correct' input data items to satisfy the required m_i criteria for that output. The Collect transaction **1000** includes an nLock-Time value (i.e. a lock time **1001**) that prevents the Collect transaction **1000** from being valid until after an agreed-upon point in time. An example of a script that would require the four pieces of data would be:

```
OP_HASH160 <H(k)> OP_EQUALVERIFY OP_DUP OP_HASH160
<H(PubA)>
OP_EQUALVERIFY OP_CHECKSIG OP_DUP OP_HASH160
<H(PubB)> OP_EQUALVERIFY
OP_CHECKSIG [TEST m_i]
```

[0151] This script is referred to as a Collect condition ([Condition: Col_i]). A 2-of-2 (m-of-n) multisig script could be utilised instead. The section [TEST m_i] acts as a representation of the script subsection that: defines the entire set of criteria that the user must satisfy, increments the counter for each criterion satisfied, and verifies that the number of criteria satisfied is correct ($\geq m_i$). This [TEST m_i] script subsection would be similar to the output script shown in $TxID_{bPay}$ and is duplicated below, where <m_i>represents the threshold for output i.

[0152] Note that the lock time restriction is not represented in the script but in the nLockTime field of the Collect transaction **1000**.

[TEST m_i]:

```
<0> OP_TOALTSTACK
[criterion_1]
  OP_IF OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_
  ENDIF
  [criterion_2]
  OP_IF OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_
  ENDIF
  ...
  [criterion_n]
  OP_IF OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_
  ENDIF
  OP_FROMALTSTACK <m_i> OP_GREATERTHANOREQUAL
```

[0153] The unlocking script (input script of the Collect transaction **1000**) would be

```
<answers> <sigB> <PubB> <sigA> <PubA> <k>
```

[0154] This script is referred to as the Collect input (<Input: Col_i>. This script contains the signatures of both participants as well as the secret value k. Also visible in the script is the value <answers>. This is the section of the input script that contains multiple data items that are expected to be the solutions that satisfy at least m_i criteria in the [TEST m_i] section of the output script of output i of the sPay transaction.

```
<answers> ≡ <input_n> <input_n-1> ... <input_2> <input_1>
```

[0155] Each of the sub elements of <answers> are to be arranged in a specific order so that each <input_i> is assessed by the intended [criterion_i] found in the [TEST] script. The element <input_i> is expected to be the recipient (Bob)'s solution to the puzzle criteria defined by the payer Alice in [criterion_i]. Note that similar to the BTP, a criterion could have sub-criteria and an input value can be composed of multiple sub inputs.

[0156] With the option of at least three different spending options (Collect, Refund, Cancel) to spend an output i of the sPay transaction, the locking (output) script for said output must incorporate the necessary script elements to facilitate the choice of spending transaction. In order to do this the locking script may makes use conditional elements, e.g. OP_IF, OP_ELSE, etc. An example of such a locking script is shown below

```
[Condition: Col_i]OP_IF (if Collect condition works ->Success)
OP_ELSE [Condition: Ref_i] OP_IF (Else try Refund. If Refund works-
>Success)
OP_ELSE [Condition: Can_i] OP_IF (Else try Cancel. If Cancel works-
>Success)
OP_ELSE 0 (if none work. Failure)
OP_ENDIF OP_ENDIF
```

[0157] Note that the text in parentheses are comments and would not actually be included in the transaction.

[0158] Here the script contains the three conditions [Condition: Col_i], [Condition: Ref_i], and [Condition: Can_i]; each governs a particular spending transaction. These conditions are the locking scripts previously described. To successfully spend the sPay output the proposed recipient Bob **103***b* must include the correct input necessary for their intended spending transaction type. The input script that decides the spending transaction type would have at least three subsections (see cell below). These inputs correspond to those previously described and are shown in cell below

```
<Input: Can_i> <Input: Ref_i> <Input: Col_i>
```

[0159] If the recipient Bob **103***b* is submitting a particular type of spending transaction e.g. Collect **1000**, then Bob **103***b* is expected to have the appropriate script subsection <Input: Col_i>be included in that three-part input script and replace the other two input elements with 'dummy data'. Represented as <dummy> for our purposes, dummy data is defined as input data that intentionally evaluates to 'incorrect' when evaluated against a [Condition]. Thus, if Bob **103***b* wants to submit the Collect transaction **1000** then the input script of the Collect transaction would resemble

```
<dummy> <dummy> <Input: Col_i>
```

[0160] FIG. **11** illustrates an example time-staggering of transactions according to the STP. A potential vulnerability of 'a transaction that has multiple outputs where each output pays based on the m_i of n criteria that a proposed recipient is able to satisfy' is captured within the statement "if a recipient knows at least m_i of n solutions, then that proposed recipient knows m_i-1 solutions, m_i-2 solutions, . . . , 2 solutions, or 1 solution." This translates to the recipient being able to spend multiple outputs of the sPay transaction. This of course is highly undesirable. Lock times and time-staggering can be used to prevent such abuse. This time-staggering may be enabled via the nLockTime field of transactions and works to ensure that the recipient can only submit a Collect transaction to only one output of the sPay transaction.

[0161] The time-staggering of the spending transactions is explained using the design shown in FIG. **11**. Time is shown on the horizontal axis, left to right. Spending transactions are shown as full-bodied arrows, where the tail of the arrow begins at the time (at or after) that the transaction may be successfully submitted to the blockchain **150**, as restricted by the lock time placed on a transaction. As an example, the Collect transaction $Collect_1$ must be submitted after time=t_1, whereas Collect transaction $Winning_2$ must be submitted after time=t_2, etc. The intended recipient of the spending transaction is shown at the head of the arrow. A Cancel transaction is not time dependent (within this context this means that the Cancel transaction's submission is not restricted by an nLockTime value). The Cancel transactions can be submitted to the blockchain **150** at any time—as long as the secret value of k (and Alice's signature) is known by the submitter. Output i=1 does not have a Cancel transaction. A Cancel transaction for any output j is utilised to give Alice **103***a* the ability to immediately 'cancel' the Collect; transaction. $Collect_j$ is only submissable after the collect transaction of an output i is submitted. Since $Collect_1$ has no 'previous' Collect, it follows that a Cancel transaction to output i=1 is inapplicable. Bearing in mind that a $Refund_1$ transaction exists for output i=1 if Bob never submits his $Collect_1$ transaction. Note that the submission of a Collect transaction to the blockchain **150** reveals the value k to any interested party.

[0162] According to FIG. **11**, the first transaction Tx_1 that may be submitted is that of $Collect_1$ of output i=1. This can be submitted by Bob after time t_1. For output i=1 the refund transaction $Refund_1$ can only be submitted (by Alice) at time $t_{1,2}$ (which is a time after time t_1 but before time t_2). Thus, after time t_1, Bob has time span $\Delta_1=t_{1,2}-t_1$ units of time to submit the Collect transaction $Collect_1$ (lest Alice refunds the output to herself). This time span is represented by the two-headed arrows.

[0163] The next Collect transaction **1000** that may be submitted is $Collect_2$ . . . which may only be submitted after time t_2. This means that if Bob **103***b* wants to collect the winnings for this second output i=1, he can only submit this transaction after t_2 which is after the time t_1,2 the time (at or after) Alice **103***a* may submit the refund transaction $Refund_i$ for output i=1.

[0164] The provides protection against Bob **103***b* collecting multiple outputs of the sPay transaction. The person funding the pay transaction sPay (Alice) has the responsibility of executing the refund transaction $Refund_1$ before the recipient has the ability to submit the Collect transaction $Collect_2$. Thus, after $t_{i,i+1}$, Alice has $\Delta_{j,i+1}=t_{i+1}-t_{i,i+1}$ units of time to submit the Refund transaction $Refund_1$ (lest Bob be able to submit $Collect_{i+1}$).

[0165] Output i=1 is an exception in that it does not have a Cancel transaction. To explore the Cancel transaction, look at a generic output i: i >1 of FIG. **11**. A Cancel transaction $Cancel_i$ performs the functionality of a refund, with the exception that, as opposed to a refund transaction $Refund_i$, the Cancel transaction's submission to the blockchain **150** is not restricted by an nLockTime value but by knowledge (or lack thereof) of the secret value. As long as the secret value is k is known, Alice **103***a* can submit this Cancel version of a 'refund' transaction. In some examples, the inclusion of Alice's signature in this transaction is also required. Given that this secret value is revealed when Bob **103***b* submits a Collect transaction, if Bob **103***b* were to submit the Collect transaction Collect; then every Cancel submission in subsequent outputs j: j >i may then immediately be submitted by Alice **103***a*. This renders all subsequent outputs as inaccessible to Bob **103***b*.

[0166] The proposed recipient Bob **103***b* thus has the responsibility to assess (off block) which of the outputs he can successfully spend and that he is willing to submit. The output selected is expected to be the one that would reward Bob with the most amount of the digital asset. If Bob **103***b* selects an output i then Bob has to keep the value k secret until he submits the transaction $Collect_i$. For any prior output (j<i) Alice would have submitted (or at least had the opportunity to) submit the Refund transaction $Refund_j$.

[0167] Whilst preferable, lock times and time-staggering are not essential. Instead, Alice **103***a* may simply submit the Cancel and/or Refund transactions at an appropriate time. Similarly, Bob **103***b* may be trusted to only submit a single Collect transaction.

[0168] FIG. **12** illustrates an example sequence for the STP, with time running from top to bottom. The sequence is as follows:

[0169] 1. Bob selects a secret value k.

[0170] 2. Bob sends a hash of the secret value to Alice.

[0171] 3. Alice creates the sPay transaction.

[0172] 4. Alice creates the Refund and Collect transactions.

[0173] 5. Alice sends the set of criteria and lock times to Bob.

[0174] 6. Alice signs the Refund transactions and Collect transactions and sends them to Bob.

[0175] 7. If Bob agrees, Bob signs the Refund transactions and sends them back to Alice.

[0176] 8. Alice submits the sPay transaction to the blockchain.

[0177] 9. Bob determines solutions to as many criteria as possible.

[0178] 10. Bob determines the output which he can unlock which pays the most amount of the digital asset.

[0179] 11. As this point, Alice may submit a Refund to the blockchain for any output j<i.

[0180] 12. Bob submits a Collect transaction for output i to the blockchain.

[0181] 13. Now that the secret value has been revealed, Alice submits Cancel transactions for

[0182] outputs j>i.

[0183] Note that the above sequence is one particular example and other orderings are possible. For instance, there is some flexibility in when the criteria are presented to Bob.

[0184] As discussed above, in some implementations of the binary and spectrum threshold protocols, successfully satisfying a criterion increments the counter by 1, or in other implementations the counter may be incremented by another value. A "weighted criteria" implementation may be e necessary in certain contexts where certain criteria are more 'important' than others. Weighted criteria may be used where the amount rewarded for satisfying a criterion is not mandatorily uniform but is dependent on what the criterion is.

[0185] Satisfying a criterion i increments the counter by w_i where $w_i \in \mathbb{N}$. The threshold is set at a value m_i: $m_i \leq v$, where $v \geq n$ is the maximum achievable (counter) value, e.g. satisfying criterion_i may increment the counter by 2, satisfying criterion_j may increment the counter by 1, whereas satisfying criterion_k may increment the counter by 3, etc.

[0186] In such implementations, arriving at the threshold is not bound only to the number of criteria satisfied, but also on what criteria are satisfied. Bob **103***b* may arrive at the threshold satisfying a varying number of criteria. As an example, if the threshold value is set as six, Bob **103***b* may achieve the threshold by satisfying three criteria where the weight of each is worth two units, or Bob **103***b* may achieve the threshold by satisfying two criteria where the weight of each is worth three units.

[0187] Both the BTP and the STP can be designed to incorporate this weighted consideration by simply changing the value that the counter held in the alt stack is incremented by in the [TEST] script to now be w_i for each criterion i. The threshold value $\langle m_i \rangle$remains the payer's choice and is chosen in consideration to the criteria and their weights.

```
<0> OP_TOALTSTACK
[criterion_1]
OP_IF OP_FROMALTSTACK <w_1> OP_ADD OP_TOALTSTACK OP_
ENDIF
[criterion_2]
OP_IF OP_FROMALTSTACK <w_2> OP_ADD OP_TOALTSTACK OP_
ENDIF
...
[criterion_n]
OP_IF OP_FROMALTSTACK <w_n> OP ADD OP_TOALTSTACK OP_
ENDIF
<m_i> OP_FROMALTSTACK OP_ GREATERTHANOREQUAL
```

[0188] For the [TEST] script shown so far (weighted and otherwise) there are improvements that can be made to improve the efficiency of its execution. The following describes at least one improvement. The efficiency improvement is made by reducing the number of input-criterion assessment (I-CA) that are made. For the previous version of the [TEST] script, each of n inputs are assessed against its corresponding criteria; this means that n I-CAs are always done. This may be problematic if the input-criterion assessment is particularly computationally expensive.

[0189] The previous version of [TEST] and the corresponding set of inputs of an unlocking script is based on the idea that the recipient Bob **103*b*** may or may not know which of the inputs would be able to satisfy the input's corresponding criterion; as such the locking script determines which are correct while updating a counter accordingly. However, it is possible that Bob **103*b*** would, in advance of submitting the Collect transaction, know which m of the n criteria he is able to satisfy. The [TEST] script can be reworked such that it asks of the Bob **103*b*** to specify which criterion it wants to perform an I-CA for.

```
<0> OP_TOALTSTACK
OP_IF [criterion_1] OP_VERIFY
OP_FROMALTSTACK <w_1> OP ADD OP_TOALTSTACK OP_ENDIF
OP_IF [criterion_2] OP_VERIFY
OP_FROMALTSTACK <w_2> OP_ADD OP_TOALTSTACK OP_ENDIF
...
OP_IF [criterion_n] OP_VERIFY
OP_FROMALTSTACK <w_n> OP ADD OP_TOALTSTACK OP_ENDIF
OP_FROMALTSTACK <m_i> OP_GREATERTHANOREQUAL
```

[0190] Note that for the revised [TEST] script an OP_IF element is introduced before each criterion. This if-statement checks an input value from Bob (value 0 or 1) to determine whether Bob wants an I-CA performed for that criterion (1 if YES and 0 if NO). The OP_IF opcode may be replaced with equivalent functions for checking for a specific data element. In general, the [TEST] script may comprise a "check" component configured to check if an input comprises one of two possible data elements. If a first data element is provided, the following script executes and vice versa for a second, different data element.

[0191] Also note the introduction of OP_VERIFY. This could be a standalone opcode after the previous elements of criterion script, or it could be that the criterion utilises verify-opcodes throughout.

Op_VERIFY after

```
OP_IF [OP_EQUAL] OP_VERIFY
```

Or VERIFY-Opcode

```
OP_IF [OP_EQUALVERIFY]
```

[0192] Note that the VERIFY Script element marks a transaction as invalid if top stack value is not true. The top stack value is also removed. If the criterion fails, the VERIFY component immediately makes the entire transaction invalid and no further script execution is conducted. In light of the revised [TEST] of the locking script, the unlocking script in the Collect transaction would also need to be revised. The unlocking script described above includes a section <answers> as shown below

```
<answers> ≡ <input_n> <input_n-1> ... <input_2> <input_1>
```

[0193] The efficiency-revised version of this section of input script would feature interspersed first or second data elements (e.g. 0 or 1 values) that indicate whether Bob wants an I-CA performed for a particular criterion. An example of such a revision is shown below

```
<answers> ≡ <input_n> <1> <0>... <0> <input_4> <1> <0> <0>
<input_1> <1>
```

[0194] Here Bob **103*b*** knows he can arrive at the required threshold by only providing solutions of criterion 1,4, and n. As such, a value of <1> is placed after the <input_1>, <input_4>, and <input_n>elements. <0>replaces every other <input_i>value that was in the <answers>script. Utilising the revised <answers> as above, the total number of I-CAs done would be 3 rather than n. As such, assuming weight <W1> is equal to 1 for all i, then only m I-CAs are done for an m-of-n threshold.

[0195] FIGS. **13** and **14** illustrate an example use-case of the efficiency-improved [TEST] script. The use case is that of an alternative to the current implementation of the m-of-n multisig functionality of some blockchain protocols. For the existing implementation of the m-of-n multisig functionality, via the OP_CHECKMULTISIG, the locking script contains a set of n public keys whereas the unlocking script contains m digital (ECDSA) signatures. The determination of whether signatures meet the threshold is done the following way. The OP_CHECKMULTISIG opcode compares the first signature against each public key until it finds an ECDSA match. Starting with the subsequent public key, it compares the second signature against each remaining public key until it finds an ECDSA match. The process is repeated until all signatures have been checked or not enough public keys remain to produce a successful result. All signatures need to match a public key. Because public keys are not checked again if they fail any signature comparison, signatures must be placed in the scriptSig using the same order as their corresponding public keys were placed in the scriptPubKey. If all signatures are valid, 1 is returned, 0 otherwise.

[0196] Consider the 3-of-5 multisig example shown in FIG. **13**. Each public key P_i can be seen as a criterion, while each signature Sig_i can be seen as an input. Bearing in mind that a dashed arrow symbolises a failed I-CA and a non-dashed arrow a successful I-CA, it can be seen that there are two I-CAs for assessing Sig_3 (similarly for Sig_3). Due to the fact that the there is no representation in the script of which public key P_i that Sig_3 is to be assessed against, Sig_3 has to be assessed, in an ordered sequence against a set of Pis until it either:

[0197] a) arrives at a matching P_i;

[0198] b) exhausts all P_i values without finding a match; or

[0199] c) exhausts enough P_i values without finding a match, such that even if it were to eventually find its match in the remaining set of P_is, there would not be enough P_is that remain that would be sufficient to meet the required threshold (even if all remaining of P_is successfully match the remaining Sig(s).

[0200] Ultimately, the key takeaway is that for OP_CHECKMULTISIG, each Sig; is involved in ≥1 PublicKey—Signature I-CAs. If the efficiency-improved [TEST] function were to be utilised there would be efficiency gains in the number of I-CAs performed. Consider such an m-of-n ECDSA signature [TEST] script as the one proposed below.

```
<0> OP_TOALTSTACK
OP_DUP OP_HASH160 P1 OP_EQUALVERIFY OP_CHECKSIG-
VERIFY
  OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_
  ENDIF
  OP_DUP OP_HASH160 P2 OP_EQUALVERIFY OP_
  CHECKSIGVERIFY
  OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_
  ...ENDIF
  OP_DUP OP_HASH160 Pn OP_EQUALVERIFY OP_
  CHECKSIGVERIFY
  OP_FROMALTSTACK <1> OP_ADD OP_TOALTSTACK OP_
  ENDIF
  OP_FROMALTSTACK <mi> OP_GREATERTHANOREQUAL
```

[0201] and its corresponding unlocking script below where Bob **103***b* provides three signatures Sig_1, Sig_4, and Sig_n,

```
<answers> ≡ < Sign >< Pn > < 1 > < 0 > ... < 0 > < Sig4 >< P4 > < 1 >
< 0 > < 0 > < Sig1 >< P1 > <1>
```

[0202] Due to the fact that Bob uses the values 0 and 1 to signify what public key a signature is assessed against, this reduces the number of I-CAs each signature is involved with to a maximum of 1 (as opposed to the ≥1 of OP_CHECKMULTISIG), as illustrated in FIG. **14**.

[0203] An additional advantage of utilizing the [TEST] function (both the previously presented version and the efficiency-improved version) is a privacy gain. By utilising the [TEST] function for m-of-n multisig instead of OP_CHECKMULTISIG, Bob **103***b* does not have to reveal any public keys. This is due to the fact that the OP_CHECKMULTISIG locking script comprises the public keys whereas the [TEST] locking script comprises the hashes of the public keys.

CONCLUSION

[0204] It will be appreciated that the above embodiments have been described by way of example only.

[0205] More generally, according to a first instantiation of the teachings disclosed herein there is provided a computer-implemented method of generating a transaction for a blockchain, the transaction being for transferring an amount of a digital asset from a first party to a second party; the method being performed by the first party and comprising: generating a first transaction comprising an output locking the amount of the digital asset, the output comprising an output script comprising a plurality of criterion components each requiring a respective input data item, and a plurality of counter script components, each criterion component being associated with one of the counter script components, and wherein the output script is configured so as to, when executed alongside an input script of a second transaction, i) increment a counter each time a respective criterion component is satisfied by a respective input data item of the input script, and ii) to require the counter to increment to at least a predetermined number in order to be unlocked by the input script.

[0206] According to a second, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the first instantiation, wherein a total number of the plurality of criterion components is greater than or equal to the predetermined number.

[0207] According to a third, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the first or second instantiations, wherein one, some or all of the plurality of criterion components define a different respective criterion.

[0208] According to a fourth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the first to third instantiations, wherein one, some or all of the plurality of criterion components comprises a plurality of sub-criterion components each requiring respective sub-input data item.

[0209] According to a fifth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the first to fourth instantiations, wherein one, some or all of the plurality of criterion components and/or one, some or all of the plurality of sub-criterion components require, as input or sub-input data items respectively, a respective blockchain signature corresponding to a predetermined blockchain public key.

[0210] According to a sixth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the first to fifth instantiations, wherein one, some or all of the plurality of criterion components are associated with a counter script component that is configured to increment the counter by a different number.

[0211] According to a seventh, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the first to sixth instantiations, wherein each input data item in the input script comprises either a first or a second element, and wherein one, some or all of the plurality of criterion components are associated with a respective check component, wherein each respective check component is configured so as to, when executed alongside the input script, i) determine whether each input data item comprises the first element, and ii) only execute the associated criterion component if the input data item comprises the first element.

[0212] According to an eighth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the seventh instantiation, wherein each output script is configured to invalidate the first transaction if an executed criterion component is not satisfied by input data of the input script.

[0213] According to a ninth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the first to eighth instantiations, comprising transmitting the respective criterion and/or sub-criterion to the second party.

[0214] According to a tenth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the first to ninth instantiations, comprising transmitting the first transaction to one or more nodes associated with the blockchain for inclusion in the blockchain.

[0215] According to an eleventh instantiation of the teachings disclosed herein there is provided computer equipment of the first party, comprising: memory comprising one or more memory units; and processing apparatus comprising one or more processing units, wherein the memory stores code arranged to run on the processing apparatus, the code being configured so as when on the processing apparatus to perform the method of any of the first to tenth instantiations.

[0216] According to a twelfth instantiation of the teachings disclosed herein there is provided a computer program embodied on computer-readable storage and configured so as, when run on computer equipment of the first party, to perform the method of any of the first to tenth instantiations.

[0217] According to a thirteenth instantiation of the teachings disclosed herein there is provided a first transaction for inclusion in a blockchain, the first transaction for transferring an amount of a digital asset from a first party to a second party, the first transaction being embodied on a computer-readable data medium or media and comprising executable code, the code comprising a plurality of criterion components each requiring a respective input data item, and a plurality of counter components, each criterion component being associated with one of the counter components, and wherein the code is configured so as to, when executed alongside an input script of a second transaction at a node of a blockchain network: increment a counter, stored in memory of the node, each time a respective criterion component is satisfied by a respective input data item of the input script; and transfer the amount of the digital asset to the second party based on the counter being incremented to at least a predetermined number.

[0218] According to a fourteenth instantiation of the teachings disclosed herein there is provided a computer-implemented method of generating a transaction for a blockchain, the transaction being for transferring an amount of a digital asset from a first party to a second party, the first and second parties each being associated with the blockchain; the method being performed by the first party and comprising: generating a first transaction comprising a plurality of outputs, each output locking a respective amount of the digital asset and comprising a respective first script condition, wherein each first script condition comprises a plurality of criterion components each requiring a respective input data item, and a plurality of counter components, each criterion component being associated with one of the counter components, and wherein each first output script condition is configured so as to, when executed alongside an input script of a second transaction, i) increment a counter each time a respective criterion component is satisfied by a respective input data item of the input script, and ii) require the counter to increment to at least a respective predetermined number in order to be unlocked by that input script, and wherein one, some or all of the first script conditions require the counter to increment to a different predetermined number.

[0219] According to a fifteenth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the fourteenth instantiation, wherein one, some or all of the outputs lock a different respective amount of the digital asset.

[0220] According to a sixteenth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the fifteenth instantiation, wherein the respective amount of the digital asset locked by a respective output is based on the respective predetermined number to which the respective first script condition requires the counter to increment.

[0221] According to a seventeenth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to sixteenth instantiations, wherein to be unlocked by the input script, each first script condition is configured to require the input script to comprise a digital signature of the first party and/or a digital signature of the second party.

[0222] According to an eighteenth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the seventeenth instantiation, comprising: for each output, generating a respective second transaction, wherein the input script of each second transaction comprises the predetermined signature of the first party; and transmitting each second transaction to the second party.

[0223] According to an nineteenth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the eighteenth instantiation, wherein each second transaction comprises a different lock time, wherein each respective lock time is configured to prevent the respective second transaction from being included in the blockchain until after a respective predetermined time.

[0224] According to a twentieth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to nineteenth instantiations, wherein each first script condition comprises a hash of a secret value, and wherein to be unlocked by the input script, each first script condition is configured to require the input script to comprise the secret value.

[0225] According to a twenty first, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to twentieth instantiations, wherein each output comprises a respective second script condition configured so as to, when executed alongside an input script of a third transaction, require the input script of the third transaction to comprise a predetermined digital signature of the first party and a predetermined digital signature of the second party in order to be unlocked by that input script.

[0226] According to a twenty second, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the twenty first instantiation, comprising: for each output, receiving a respective third transaction from the second party, wherein the input script of each third transaction comprises the predetermined signature of the first party and the predetermined digital signature of the second party.

[0227] For example, the first party may sign the third transaction and transmit the signed third transactions to the second party. The second party may then transmit the third transaction to the first party.

[0228] According to a twenty third, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the twenty second instantiation, wherein each third transaction comprises a different lock time, wherein each respective lock time is configured to prevent the respective second transaction from being included in the blockchain until after a respective predetermined time.

[0229] According to a twenty fourth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the twenty third instantiation when dependent on at least the nineteenth instantiation, wherein each output is associated with a different one of the second transactions and a different one of the third transactions, and wherein for each output, the respective lock time of the third transaction prevents the third transaction from being included in the blockchain until a time later than a time at which the second transaction can be included in the blockchain.

[0230] According to a twenty fifth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to twenty fourth instantiations, wherein one, some or all of the outputs comprise a respective third script condition, wherein each third script condition is configured so as to, when executed alongside an input script of a fourth transaction, require the input script of the fourth transaction to comprise a predetermined digital signature of the first party in order to be unlocked by that input script.

[0231] According to a twenty sixth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the twenty fifth instantiation when dependent on at least the twentieth instantiation, wherein each third script condition comprises the hash of the secret value, and wherein to be unlocked by the input script of the fourth transaction, each third script condition is configured to require that input script to comprise the secret value.

[0232] According to a twenty seventh, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the twenty sixth instantiation, comprising: obtaining the secret value; for one, some or all of the outputs, generating a respective fourth transaction, wherein the input script of each fourth transaction comprises the predetermined signature of the first party and the secret value; and transmitting each fourth transaction to one or more nodes associated with the blockchain for inclusion in the blockchain.

[0233] According to a twenty eighth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to twenty seventh instantiations, wherein a total number of the plurality of criterion components is greater than each of the respective predetermined numbers.

[0234] According to a twenty ninth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to twenty eighth instantiations, wherein one, some or all of the plurality of criterion components define a different respective criterion.

[0235] According to a thirtieth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to twenty ninth instantiations, wherein one, some or all of the plurality of criterion components comprises a plurality of sub-criterion components each requiring respective sub-input data item.

[0236] According to a thirty first, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to thirtieth instantiations, wherein for one, some or all of the first script conditions, one, some or all of the plurality of criterion components and/or one, some or all of the plurality of sub-criterion components require, as input or sub-input data items respectively, a respective blockchain signature corresponding to a predetermined blockchain public key.

[0237] According to a thirty second, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to thirty first instantiations, wherein for one, some or all of the first script conditions, one, some or all of the plurality of criterion components are associated with a counter script component that is configured to increment the counter by a different number.

[0238] According to a thirty third, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to thirty second instantiations, wherein each input data item in the input script comprises either a first or a second element, and wherein one, some or all of the plurality of criterion components are associated with a respective check component, wherein each respective check component is configured so as to, when executed alongside the input script, i) determine whether each input data item comprises the first element, and ii) only execute the criterion script components corresponding to input data items comprising the first element.

[0239] According to a thirty fourth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the thirty third instantiation, wherein each first script condition is configured to invalidate the first transaction if an executed criterion script is not satisfied by input data of the input script.

[0240] According to a thirty fifth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to thirty fourth instantiations, comprising transmitting the respective criterions and/or sub-criterions to the second party.

[0241] According to a thirty sixth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the fourteenth to thirty fifth instantiations, comprising transmitting the first transaction to one or more nodes associated with the blockchain for inclusion in the blockchain.

[0242] According to a thirty seventh instantiation of the teachings disclosed herein, there is provided computer equipment of the first party, comprising: memory comprising one or more memory units; and processing apparatus comprising one or more processing units, wherein the memory stores code arranged to run on the processing apparatus, the code being configured so as when on the processing apparatus to perform the method of any of the fourteenth to thirty sixth instantiations.

[0243] According to a thirty eighth instantiation of the teachings disclosed herein, there is provided a computer program embodied on computer-readable storage and configured so as, when run on computer equipment of the first party, to perform the method of any of the fourteenth to thirty sixth instantiations.

[0244] According to a thirty ninth instantiation of the teachings disclosed herein, there is provided a computer-implemented method of generating a transaction for a blockchain, the transaction being for transferring an amount of a digital asset from a first party to a second party, the first and second parties each being associated with the blockchain, wherein the blockchain comprises a first transaction comprising an output locking the amount of the digital asset and comprising a first script condition, wherein the first script condition comprises a plurality of criterion components each requiring a respective input data item; the method being performed by the second party and comprising: generating a second transaction comprising a first input script component, the first input script component comprising: i) a plurality of input data items each corresponding to a respective criterion component of the first script condition, ii) a secret value, iii) a predetermined digital signature of the first party, and iv) a predetermined signature of the second party; and transmitting the second transaction to one or more nodes associated with the blockchain for inclusion in the blockchain.

[0245] According to a fortieth, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the thirty ninth instantiation, wherein the first transaction comprises second and third script conditions, and wherein the second transaction comprises: a second input script component configured to evaluate as false when executed against the second script condition; and a third input script component configured to evaluate as false when executed against the third script condition.

[0246] According to a forty first, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with the thirty ninth or fortieth instantiations, wherein the first script condition comprises a plurality of check components, each associated with a respective one of the plurality of criterion components, and wherein each respective check component is configured so as to, when executed alongside the input script, i) determine whether each input data item comprises the first element, and ii) only execute the criterion script components corresponding to input data items comprising the first element; and wherein each input data item comprises either the first data element or a second, different data element.

[0247] According to a forty second, optional instantiation of the teachings disclosed herein, there is provided a method in accordance with any of the thirty ninth to forty first instantiations, wherein one, some or all of the input data items comprises a respective digital signature.

[0248] According to a forty third instantiation of the teachings disclosed herein, there is provided computer equipment of the second party, comprising: memory comprising one or more memory units; and processing apparatus comprising one or more processing units, wherein the memory stores code arranged to run on the processing apparatus, the code being configured so as when on the processing apparatus to perform the method of any of the thirty ninth to forty second instantiations.

[0249] According to a forty fourth instantiation of the teachings disclosed herein, there is provided a computer program embodied on computer-readable storage and configured so as, when run on computer equipment of the second party, to perform the method of any of the thirty ninth to forty second instantiations.

[0250] According to a forty fifth instantiation of the teachings disclosed herein, there is provided a first transaction for inclusion in a blockchain, the first transaction for transferring an amount of a digital asset from a first party to a second party, the first transaction being embodied on a computer-readable data medium or media and comprising executable code, the code comprising a plurality of output scripts, each output script locking a respective amount of the digital asset and comprising respective first, second, and third script conditions, each first script condition comprising i) a plurality of criterion components each requiring a respective input data item, and ii) a plurality of counter components, each criterion component being associated with one of the counter components, and wherein the code is configured so as to, when a respective output script is executed alongside an input script of a second transaction at a node of a blockchain network: if the respective first script condition is executed, increment a counter, stored in memory of the node, each time a respective criterion component is satisfied by a respective input data item of the input script, and transfer the amount of the digital asset to the second party based on the counter being incremented to at least a predetermined number; if the respective second script condition is executed, determine if the input script comprises a predetermined digital signature of the first party and a predetermined digital signature of the second party, and transfer the amount of the digital asset to the first party; and if the respective third script condition is executed, determine if the input script comprises the predetermined digital signature of the first party and a secret value, and transfer the amount of the digital asset to the first party.

[0251] According to a forty sixth instantiation of the teachings disclosed herein, there is provided a computer-readable storage medium having stored thereon the first transaction of the forty fifth instantiation.

[0252] According to a forty seventh instantiation of the teachings disclosed herein, there is provided a computer-readable storage medium having stored thereon the first transaction of the thirteenth instantiation.

[0253] According to another instantiation of the teachings disclosed herein, there may be provided a method comprising the actions of the first party and the second party.

[0254] According to another instantiation of the teachings disclosed herein, there may be provided a system comprising the computer equipment of the first party and the second party.

[0255] According to another instantiation of the teachings disclosed herein, there may be provided a set of transactions comprising the first and second transactions.

[0256] Other variants or use cases of the disclosed techniques may become apparent to the person skilled in the art once given the disclosure herein. The scope of the disclosure is not limited by the described embodiments but only by the accompanying claims.

1. A computer-implemented method of generating a transaction for a blockchain, the method being performed by computer equipment of a first party and comprising

constructing a locking script of a first blockchain transaction, wherein the locking script comprises a plurality of criterion components, a plurality of counter script components, and a predetermined number, each criterion component being linked with one of the counter script components, each criterion component comprising one or more functions, and each counter script component comprising one or more functions;

using the locking script to lock a first output of the first blockchain transaction;

configuring the locking script so that, when executed by a blockchain node together with an unlocking script of a second blockchain transaction, the locking script causes the second blockchain node to increment a counter, stored in memory of the blockchain node, each time the one or more functions of a respective criterion component determine that the input script comprises a respective required data item; and

configuring the locking script so that, when executed by the blockchain node together with the unlocking script of the second blockchain transaction, the locking script causes the second blockchain node to verify that the counter has incremented to at least the predetermined number in order for the first output to be unlocked by the unlocking script.

2. The method of claim **1**, comprising transmitting the first transaction to one or more blockchain nodes.

3. The method of claim **1**, wherein a total number of the plurality of criterion components is greater than or equal to the predetermined number.

4. The method of claim **1**, wherein one, some or all of the plurality of criterion components define a different respective criterion.

5. The method of claim **1**, wherein one, some or all of the plurality of criterion components comprises a plurality of sub-criterion components each requiring respective sub-input data item.

6. The method of claim **5**, wherein one, some or all of the plurality of criterion components and/or one, some or all of the plurality of sub-criterion components require, as input or sub-input data items respectively, a respective blockchain signature corresponding to a predetermined blockchain public key.

7. The method of claim **1**, wherein one, some or all of the plurality of criterion components are linked with a counter script component that is configured to increment the counter by a different number.

8. The method of claim **1**, wherein each input data item in the unlocking script comprises either a first or a second element, and wherein one, some or all of the plurality of criterion components are associated with a respective check component, wherein each respective check component is configured so as to, when executed alongside the unlocking script, i) determine whether each input data item comprises the first element, and ii) only execute the associated criterion component if the input data item comprises the first element.

9. The method of claim **8**, wherein each locking script is configured to invalidate the first blockchain transaction if an executed criterion component is not satisfied by input data of the input script.

10. A non-transitory computer-readable storage medium comprising a computer program configured so as, when run on computer equipment, the computer equipment performs a computer-implemented method of generating a transaction for a blockchain, the method being performed by computer equipment of a first party and comprising:

constructing a locking script of a first blockchain transaction, wherein the locking script comprises a plurality of criterion components, a plurality of counter script components, and a predetermined number, each criterion component being linked with one of the counter script components, each criterion component comprising one or more functions, and each counter script component comprising one or more functions;

using the locking script to lock a first output of the first blockchain transaction;

configuring the locking script so that, when executed by a blockchain node together with an unlocking script of a second blockchain transaction, the locking script causes the second blockchain node to increment a counter, stored in memory of the blockchain node, each time the one or more functions of a respective criterion component determine that the input script comprises a respective required data item; and

configuring the locking script so that, when executed by the blockchain node together with the unlocking script of the second blockchain transaction, the locking script causes the second blockchain node to verify that the counter has incremented to at least the predetermined number in order for the first output to be unlocked by the unlocking script.

11. A computer-implemented method of generating a transaction for a blockchain, wherein the blockchain comprises a first blockchain transaction comprising a first output locked by a locking script, wherein the locking script comprises a plurality of criterion components, a plurality of counter script components, and a predetermined number, each criterion component being linked with one of the counter script components, each criterion component comprising one or more functions, and each counter script component comprising one or more functions, wherein the locking script is configured so that, when executed by a blockchain node together with an unlocking script of a second blockchain transaction, the locking script causes the second blockchain node to increment a counter, stored in memory of the blockchain node, each time the one or more functions of a respective criterion component determine that the input script comprises a respective required data item, wherein the locking script is configured so that, when executed by the blockchain node together with the unlocking script of the second blockchain transaction, the locking script causes the second blockchain node to verify that the counter has incremented to at least the predetermined number in order for the first output to be unlocked by the unlocking script; the method being performed by computer equipment of a second party and comprising:

determining a plurality of respective required data items;

constructing an unlocking script comprising the plurality of respective required data items;

generating a second blockchain transaction comprising a first input, the first input comprising a reference to the first output of the first blockchain transaction and the unlocking script; and

transmitting the second transaction to one or more nodes associated with the blockchain for inclusion in the blockchain.

12. The method of claim **11**, comprising including a secret value in the unlocking script.

13. The method of claim **11**, comprising:

receiving a first digital signature from the first party; and

including the first digital signature of the first party in the unlocking script.

14. The method of claim **11**, comprising:

generating a second digital signature using a private key controlled by the second party; and including the second digital signature in the unlocking script.

15. The method of claim **11**, wherein one, some or all of the required data items comprises a respective digital signature.

16. A non-transitory computer-readable storage medium comprising a computer program configured so as, when run on computer equipment, the computer equipment performs a computer-implemented method of generating a transaction for a blockchain, wherein the blockchain comprises a first blockchain transaction comprising a first output locked by a locking script, wherein the locking script comprises a plurality of criterion components, a plurality of counter script components, and a predetermined number, each criterion component being linked with one of the counter script components, each criterion component comprising one or more functions, and each counter script component comprising one or more functions, wherein the locking script is configured so that, when executed by a blockchain node together with an unlocking script of a second blockchain transaction, the locking script causes the second blockchain node to increment a counter, stored in memory of the blockchain node, each time the one or more functions of a respective criterion component determine that the input script comprises a respective required data item, wherein the locking script is configured so that, when executed by the blockchain node together with the unlocking script of the second blockchain transaction, the locking script causes the second blockchain node to verify that the counter has incremented to at least the predetermined number in order for the first output to be unlocked by the unlocking script; the method being performed by computer equipment of a second party and comprising:

determining a plurality of respective required data items;

constructing an unlocking script comprising the plurality of respective required data items;

generating a second blockchain transaction comprising a first input, the first input comprising a reference to the first output of the first blockchain transaction and the unlocking script; and

transmitting the second transaction to one or more nodes associated with the blockchain for inclusion in the blockchain.

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